Professional Project

Entrepreneurship - Module

“ rAsif ”

The Smart Transport Hub

« Know the When, Master the Day»

Prepared by*:* Israa Bazzal 103809, Fatima Somaka 102212, Rana Ali Bourji 100975, Sarah Atat 104025

Under the direction of: Dr. Rawan Raya

Academic year 2025 - 2026

Table Of Contents

Executive Summary

1. **Business Idea / Product or Service**
   rAsif is a smart mobility startup offering an integrated transportation solution for university students through smart bus stations, real-time GPS tracking, arrival prediction technology, and a centralized digital platform for management and analytics. The core value of rAsif lies in providing students with precise bus arrival information, reducing waiting time, uncertainty, and safety risks. The system also equips institutions with tools to optimize fleet utilization, lower operational costs, and generate revenue through targeted digital advertising.
2. **Target Market / Customers**
   Our primary customers include university students who rely on daily transportation, drivers and fleet operators needing streamlined route management, universities and transport administrators seeking efficient solutions, and local businesses aiming to reach student communities. The initial focus will be on Lebanese universities with high commuting volumes, with plans to scale into similar regional markets facing fragmented public transport.
3. **Type of Organization / Business Model**
   rAsif operates as a technology service provider (B2B/B2B2C), combining hardware sales (smart stations with LED panels and sensors), recurring software subscriptions for universities and operators, and digital advertising revenue within the platform. This blended business model creates a diversified, scalable, and high-margin revenue stream.
4. **Owners**
   The business is founded by four university students: Rana, Sarah, Israa, and Fatima. The name "rAsif" reflects both the founders' initials and the Arabic word "رصيف," meaning platform or bus station, symbolizing the mission to modernize mobility.
5. **Industry Outlook**
   The smart mobility and intelligent transportation industry is rapidly expanding due to urbanization, sustainability goals, and growing demand for real-time solutions. In Lebanon, the lack of structured transport systems creates an urgent need for low-cost, tech-enabled mobility solutions, giving rAsif a strong early-mover advantage. As adoption increases, growth is expected through partnerships, advertising, and regional expansion.
6. **Management Skills and Resources**
   The founding team combines engineering and technical development, UI/UX and product design, business development, marketing strategy, research, and project management. Future growth will require additional expertise in software engineering (AI, backend, mobile), financial management, partnerships, and hardware manufacturing/vendor agreements. Early-stage operations will rely on a lean, cross-functional team with selective outsourcing support.
7. **Marketing and Sales Strategy**
   Marketing efforts focus on university partnerships and pilot programs, targeted digital campaigns on social media and search engines, collaboration with student clubs and campus ambassadors, and local business advertising packages. The messaging emphasizes smart, reliable, student-centered mobility that saves time and reduces stress. Sales are primarily relationship-driven, targeting universities through demonstrations, trials, and long-term service contracts.
8. **Short- and Long-Term Business Goals**
   Short-term goals (Year 1–2) include deploying pilots in 1–2 universities, validating demand and user adoption, developing a scalable platform, securing funding and partnerships, and generating initial recurring revenues. Long-term goals (Year 3–5) include expanding nationwide to major universities, introducing premium analytics tools, scaling the advertising network, expanding regionally to MENA universities, and achieving sustainable profitability.
9. **Financial Projections and Loan Request**
   Startup capital will fund hardware prototyping, software development and hosting, marketing campaigns, pilot programs, and operational costs. Initial funding needs are estimated at $80,000–

120,000 to reach MVP, pilot deployment, and first commercial contracts. Based on projected adoption, Year 1 will focus on development with low revenues and net loss, Year 2 on pilot deployments with first contracts and small advertising revenue, Year 3 reaching 5–7 institutions and break-even, and Years 4–5 achieving national coverage and profitability driven by subscription fees, hardware sales, and advertising income. We are seeking a $100,000 loan to cover hardware prototypes ($25,000), AI-driven software development ($45,000), and marketing, pilots, and operations ($30,000). This funding enables commercial readiness within 12 months, pilot deployments generating revenue contracts, and a scalable, defensible market position. Repayment is expected to start from Year 3, supported by recurring SaaS subscriptions, national university partnerships, and advertising monetization, ensuring stable cash flow and investor returns.

General overview : Industry and Startup

The Smart Transport Hub operates within the **public transportation**, **smart mobility**, and **smart city technologies** industries. These sectors focus on the digital transformation of transportation systems through tools such as GPS tracking, Internet of Things (IoT) devices, cloud platforms, artificial intelligence, and real-time information interfaces. Within this industry, our startup positions itself at the intersection of **intelligent transportation systems (ITS)** and **digital mobility infrastructure**, offering integrated solutions such as smart LED stations, real-time passenger information, predictive analytics, and mobile tools for fleet monitoring. The purpose of this industry is to improve the efficiency, reliability, and coordination of public transportation—making it more predictable, sustainable, and user-friendly.

These industries are currently experiencing strong global growth. The **global public transportation market** was valued at **$223.8 billion in 2023** and is expected to reach **$374.1 billion by 2030**, with an annual growth rate (CAGR) of **7.6%**, reflecting rising urbanization and increased reliance on mass transit systems [1] (Grand View Research, 2023). Similarly, the **smart transportation market**, which includes digital real-time tracking, AI-driven ETA forecasting, and advanced mobility analytics, is estimated at **$33.4 billion in 2024** and projected to reach **$46.4 billion by 2029**, growing at **6.8% CAGR** [2] (WIPO – Future of Transportation, 2024). On a broader scale, the **transit and ground passenger transport sector** is forecasted to expand from **$706.2 billion in 2025** to **$1.48 trillion by 2034**, driven by digitization and smart infrastructure adoption, with an estimated **8.7% CAGR** [3] (GlobeNewsWire, 2025). These statistics confirm that smart mobility is a high-growth industry with increasing investment and global demand.

The Smart Transport Hub is a **market-pull innovation**, developed in response to a clear unmet need in the Lebanese market—particularly in regions such as the Bekaa. Students and daily commuters frequently experience unreliable transportation, long and unpredictable waiting times, overcrowded pickup points, and a complete lack of visibility regarding bus or van arrival schedules. Furthermore, universities, transportation companies, and local municipalities lack digital tools that could allow them to monitor fleets, manage routes efficiently, or communicate delays to passengers. These problems create inefficiency, frustration, safety concerns, and lost productivity. The strong demand for a structured, real-time, technology-supported transport solution makes this project highly relevant for the Lebanese context.

To address this need, the Smart Transport Hub introduces a **connected smart mobility ecosystem** composed of three main components. First, **smart LED/LCD stations** installed at campuses and high-traffic stops display real-time arrival information, route details, delay notifications, and advertisements. Second, a **driver mobile application** enables transport operators to activate real-time GPS tracking using their smartphones, eliminating the need for costly hardware installations. Third, an **AI-powered prediction platform** processes the GPS data to calculate estimated arrival times (ETA), detect unusual delays or route changes, and update the station interfaces instantly. Together, these components create a comprehensive, scalable, and low-cost digital transport infrastructure suitable for university fleets and community transport networks.

Although transportation apps do exist in Lebanon, they predominantly focus on ride-hailing and private car mobility. They do not support fixed-route buses, do not offer public-facing smart station displays, and lack AI-based predictive capabilities. In contrast, the Smart Transport Hub provides a localized, data-driven, and operationally optimized solution specifically designed for universities and commuter transport. This creates a strong competitive advantage, particularly as the Bekaa region currently has no comparable system in place. Additionally, the platform enables new revenue streams through digital advertising displayed on the smart stations, making the model financially sustainable.

Mission Statement:

"To make daily transportation reliable, transparent, and efficient by delivering smart, real-time mobility solutions that empower students, commuters, and transport operators

(1) Grand View Research (2023) – Public Transportation Market Size Report
(2) WIPO (2024) – Future of Transportation – Technology Trends
(3) GlobeNewsWire (2025) – Transit and Ground Passenger Transport Industry Forecast

3.1 Market Research and External factors analysis

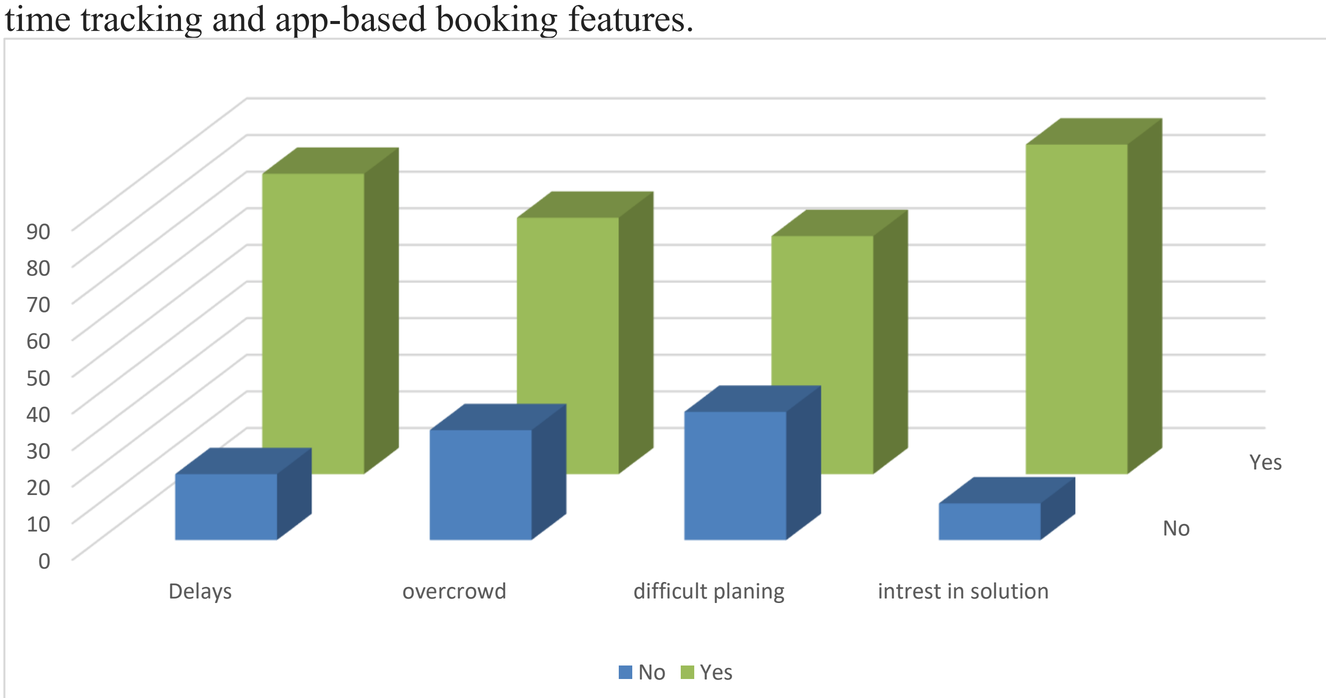
3.1.1. Primary Market Research

Methodology:

We conducted a survey (https://docs.google.com/forms/d/e/1FAIpQLSeMCHAuTigQh1PPZXTrebgDddzc0AUBrnTJeunxGrexPAziSA/viewform?usp=dialog) targeting students and daily commuters in the Bekaa region to understand their public transportation experiences, challenges, and willingness to use smart transport solutions. The survey focused on travel habits, waiting times, awareness of schedules, and interest in real-time tracking and booking services.

Key Findings:

- Frequent delays: Majority of respondents reported long and unpredictable waiting times for buses and vans.
- Overcrowding: Many students experienced overcrowded stops, especially during peak hours.
- Planning difficulties: Commuters expressed difficulty organizing their trips due to a lack of reliable information.
- Interest in smart solutions : A high percentage of participants showed willingness to use real-time tracking and app-based booking features.



Comments:

The survey confirms a strong demand for reliable, tech-enabled transportation. Students and commuters are receptive to digital solutions that improve efficiency, reduce waiting time, and provide accurate arrival information.

3.1.2. Secondary Market Research

1. Description o f the Secondary Market Research

Secondary market research was conducted using existing studies, global industry reports, Lebanese transportation studies, and analyses of international and local transport systems.
The objective was to understand:

- Global trends in smart mobility and intelligent transportation systems (ITS)
- Transportation challenges faced by students in Lebanon, especially in the Bekaa
- Existing solutions used internationally and locally
- Gaps and opportunities in the current market
- External factors affecting feasibility and market readiness

This research relied entirely on already published data—market reports, academic studies, government publications, and competitor platforms—to validate the relevance and viability of the Smart Transport Hub.

2. Results of the Secondary Market Research

A. Global Market Findings

- The smart mobility and intelligent transportation market is experiencing **rapid global growth**, projected to exceed **$150 billion by 2030**, with an annual growth rate between **10–14%**.[1]
- There is increasing global adoption of **AI-based ETA prediction**, **GPS-enabled fleet tracking**, and **smart station displays**.[2]
- International systems such as **NextBus**, **Citymapper**, and **Dubai Smart Bus Stops** demonstrate that real-time passenger information systems are becoming standard in modern cities.[3]

B. Lebanon & Bekaa Transport Findings

- Studies on Lebanese public transport show **high unreliability**, lack of schedules, and strong dependency on informal vans and buses[4].
- Students in the Bekaa experience: [5]
  - Long waiting times
  - Overcrowded pick-up points
  - No real-time information
  - Difficulty planning trips
- No university or municipality in Bekaa currently uses smart stations, digital displays, or AI-powered ETA systems.

(1) Grand View Research – global smart mobility statistics.
(2) GlobeNewswire/PRN – ITS market growth forecasts.
(3) International systems review (NextBus, Citymapper, RTA Dubai).
(4) World Bank – Lebanese transport infrastructure data.
(5) UN-Habitat – Lebanon mobility challenges.

C. Competitor & Market Gap Findings

- **Local apps** (Bolt, Allo Taxi, Careem):[6]
    - Focus only on ride-hailing
    - Do **not** support fixed-route buses or stations
    - Do **not** offer predictive analytics or physical displays
- **International solutions** exist but are designed for large urban systems—not student commuting.
- There is a **clear market gap** for a real-time, affordable, student-focused smart station system.

D. External Market & Feasibility Findings (from PESTEL & Industry Data)

- Political and economic conditions create challenges (inflation, currency devaluation).
- Technological environment is favorable due to low-cost GPS and improved AI tools.
- Social conditions support adoption, as Lebanese youth are highly tech-oriented.
- Legal requirements involve permits and data privacy compliance but are manageable.

3. Sources Used

(Full citations are included in the Appendices)

- Grand View Research (Smart Transportation Market Report, 2023)
- GlobeNewswire (Transit Market Forecast, 2025)
- Statista / MarketWatch (Smart Mobility Growth, 2024)
- World Bank — *Lebanon Urban Transport Technical Report* (2023)
- UN-Habitat — *Mobility Challenges in Lebanon* (2022)
- International systems: NextBus, Citymapper, Dubai RTA Smart Bus Shelters
- Local competitor apps: Bolt, Allo Taxi, Careem

These sources collectively validate global demand, local challenges, and the opportunity for an intelligent transport solution in the Bekaa region.

4. Analysis of the Data

Based on the collected information:

- Market growth trends confirm long-term sustainability and financial potential.
- The Bekaa region shows **urgent unmet needs**—unreliable transportation, no real-time visibility, and no existing smart infrastructure.
- Comparing Lebanon's systems with international models shows that **physical smart stations + AI prediction** are completely absent locally.
- Competitor analysis reveals that **no app or system** currently serves student commuters or fixed-route vans.
- External macro factors (PESTEL) highlight risks but also major opportunities for innovation.

(6) Local competitor analysis (Bolt, Allo Taxi, Careem).

5. Comments / Interpretation

The secondary research strongly supports the project concept:

- There is a **significant market gap** and unmet need in the Bekaa region for a real-time transport information system.
- Global trends prove the model is modern, scalable, and aligned with the future of mobility.
- The solution is **feasible**, especially due to affordable GPS and AI technologies.
- Universities and transport operators can benefit directly from reduced chaos and improved passenger communication.
- Monetization through advertising and subscriptions is validated by global smart station models.

Overall, secondary research confirms that the Smart Transport Hub is necessary, feasible, and supported by both global trends and local market needs.

3.1.3. PESTEL Analysis

The environmental influences on the company into six main macro-environmental categories: Political, Economic, Sociological, Technological, Ecological and Legal.

Category	Possible factors	Business impact	Impact Opportunity /Threat	Time frame
Political	Ongoing Israeli-Lebanese Tensions	Potential delays in infrastructure development due to security concerns.	Threat	Unknown
	Fragile Political Stability	Risk of policy reversals or project delays.	Threat	Short Term
	Government's Reform Agenda	Possible alignment with smart infrastructure initiatives.	Opportunity	Medium Term
Economic	Currency Devaluation & Inflation	Increased costs for imported technology and materials.	Threat	Short to Medium Term
	Limited Access to Capital	Challenges in securing funding for large-scale projects.	Threat	Short to Medium Term
	World Bank Reconstruction Efforts	Potential for funding and support for infrastructure projects.	Opportunity	Medium Term
Social/Sociocultural	High dependence of students on public transportation	increase demand for smart and reliable transport systems	opportunity	Medeuim term

	Growing acceptance of digital and smart solutions among youth	Encourages quick adoptation of the smart station and app	opportiniy	Short term
	Resistance from some traditional drivers or transport companies	May delay cooperation and integration	threat	Short term
Technological	Availability of affordable GPS and IoT technologies	Enables real-time tracking at low cost	opportiniy	Short term
	Advances in AI and data analytics	Improve accuracy of bus arrival predictions	opportunity	Medeuim to long term
	Limited internet coverage in some rural areas	May effect system reliability	threat	Short term
Legal	Need for government or municipal authorization to install public smart stations	May delay project approval or deployment	threat	Short to medium te4rm
	Data privacy and GPS tracking regulation	Require compliance with Lebanese digital data laws(privacy,consent for tracking)	Thrat	Medium Term
	Intellectual property and software licensing	Protects the system's unique software and brand identity	Opportunity	Long Term
Ecologic	Use of energy-efficient LED screens	Reduces electricity consumption and operational costs	Opportunity	Short Term
	Potential need for recycling old electronic parts or screens	May require proper disposal prosedures	Threat	Medium Term
	Support for eco-friendly public transportation initiatives	Aligns with global sustainability goals and may affect funding	Opportunity	Medium tp Long Term

PESTEL Analysis – Study Result

Category	Years				
	1	2	3	4	5
Political	Highly risky	Highly risky	risky	Not risky	Good
Economic	Highly risky	Highly risky	risky	Not risky	Good
Social/Sociocultural	Not risky	Good	Good	Good	Good
Technological	Good	Good	Good	Good	Good
Legal	risky	risky	Not risky	Good	Good

3.1.4.Industry Analysis

A. Industry or set of industries :

The Smart Transport Hub operates within the Smart Mobility and Intelligent Transportation industry, which combines digital infrastructure, AI analytics, and IoT technologies to optimize public and urban transport.

- **NAICS Codes:**
  - 485113 — Bus and Other Transit Systems
  - 541511 — Custom Computer Programming Services (for the AI and app software)
  - 541512 — Computer Systems Design Services (integration of software and smart stations)
  - 423430 — Computer and Computer Peripheral Equipment and Software Merchant Wholesalers (for LED screens and hardware)

B. Factors that influence the demand for its product or service

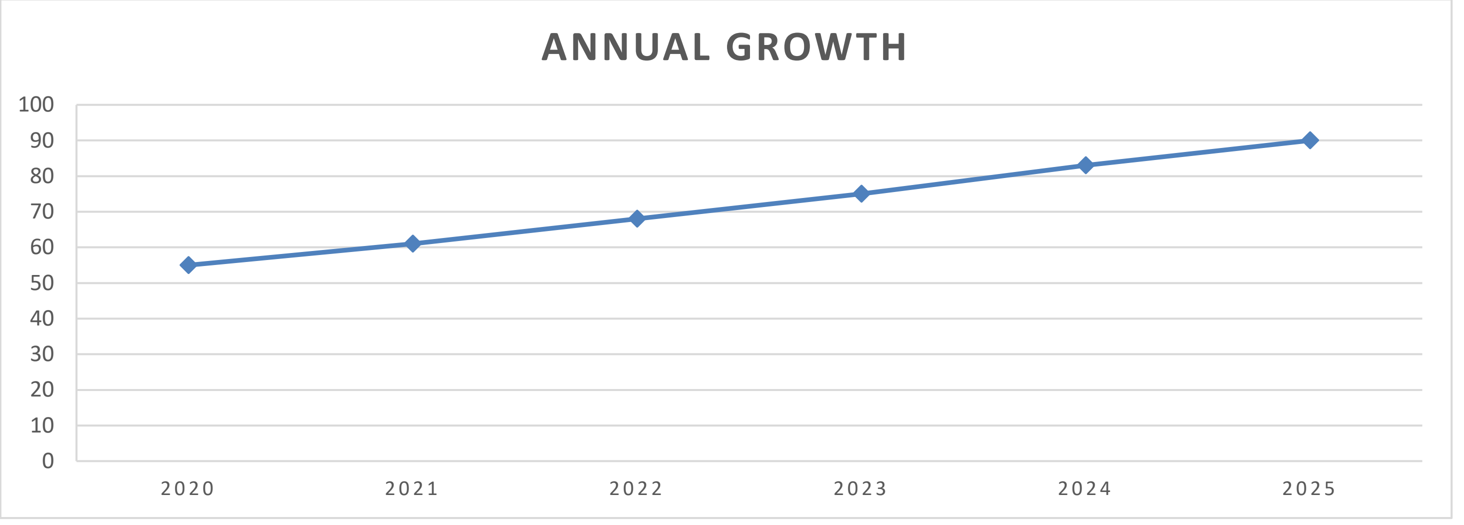
- **Urbanization and population growth:** More students and commuters create higher demand for organized transport.
- **Public dissatisfaction with existing transport:** Delays, overcrowding, and poor scheduling increase willingness to adopt smart solutions.
- **Adoption of digital solutions:** Younger populations are increasingly comfortable using apps for real-time tracking and seat reservations.
- **Economic factors:** Affordability and subscription pricing affect adoption rates among students.
- **Government initiatives:** Policies promoting smart city infrastructure or public transport improvements can encourage demand.

C. Factors that influence the supply for its product or service

- **Availability of hardware and technology:** LED screens, GPS modules, cloud services, and IoT devices are essential.
- **Skilled workforce:** AI developers, app designers, and system integrators are required to implement and maintain the system.
- **Partnerships with transport companies and municipalities:** Supply depends on cooperation with drivers, universities, and transport operators.
- **Internet and electricity reliability:** Stable connectivity and power are necessary for continuous operation.
- **Costs of installation and maintenance:** Hardware procurement and maintenance influence supply capabilities.

D. Industry size (historic, current, projected)

- **Historic (2020–2025):** The global smart mobility market grew from approximately $55B in 2020 to $90B in 2025, reflecting increasing adoption of intelligent transport systems and digital infrastructure.



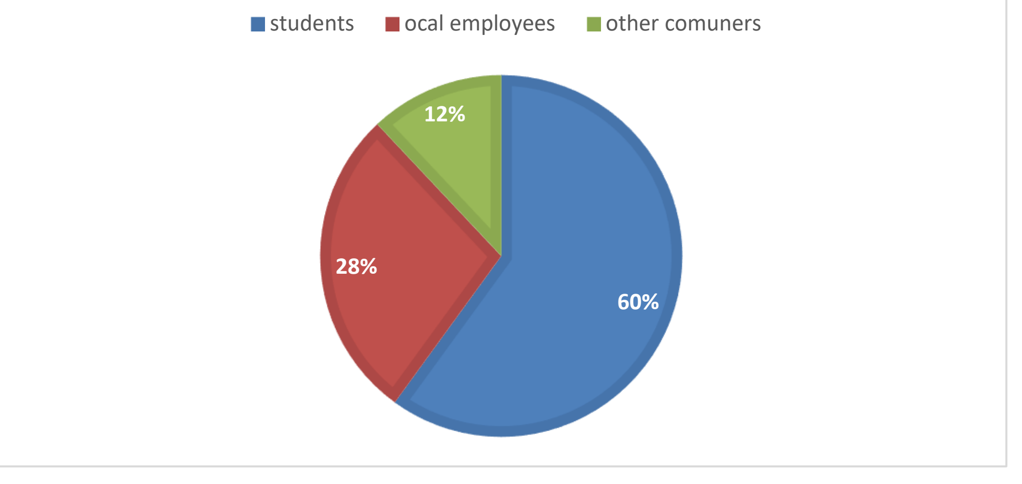
- **Current (2025):** ~$90B globally, with Lebanon in early stages of adoption; Bekaa represents a small but underserved segment with strong growth potential.
- **Projected (2030):** $150B globally, ~10–14% annual growth, driven by urbanization, smart city initiatives, and AI integration.

E. Current and anticipated industry characteristics and trends

- **Current:**
  - Focus on real-time tracking and data analytics
  - Integration of GPS and IoT for fleet management
  - Use of apps for user engagement and seat reservations
- **Anticipated Trends:**
  - Wider adoption of AI-based predictive systems
  - Expansion of smart transport hubs to smaller cities and university campuses
  - Increased digital advertising and monetization through smart stations
  - Growth of subscription-based models for premium transport features

F. Major customer groups for the industry (consumers, governments, businesses)

- **Consumers:** Students, daily commuters, and employees relying on public transport.

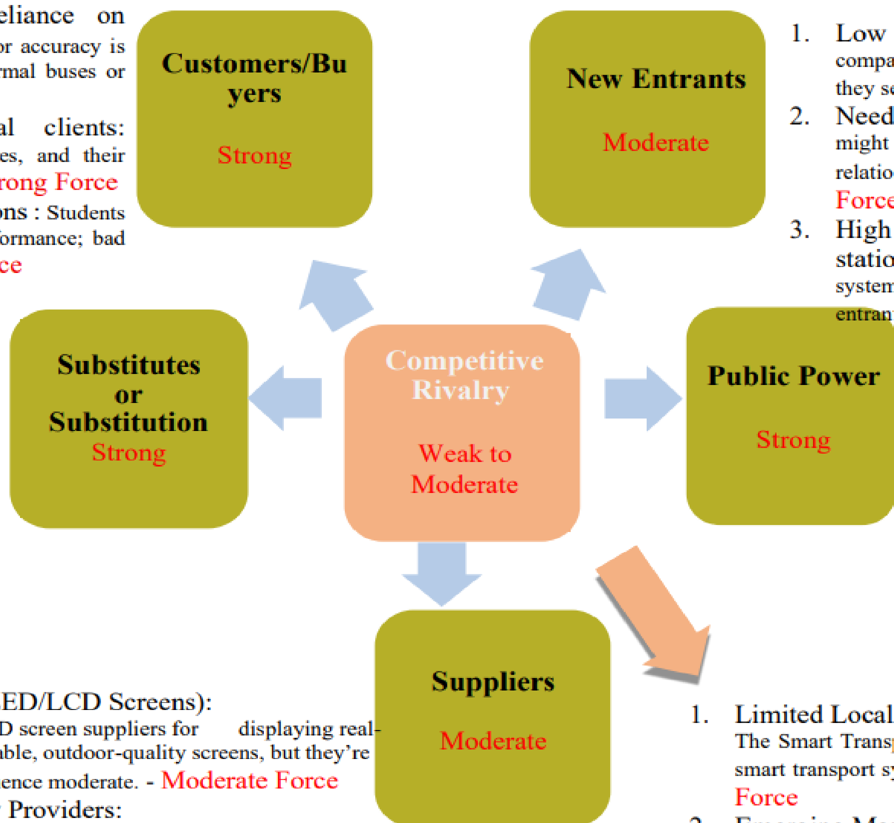


- **Businesses:** Public and private transport operators, universities, local municipalities.
- **Government:** Ministry of Transport and municipal authorities for permitting and integration with public transport initiatives.

G. Size of the target market (number of customers, size of purchases, frequency of purchases, trends

- **Number of customers:** In Bekaa, ~15,000–20,000 students commute daily; local employees add another ~5,000–10,000 potential users.
- **Size of purchases:** Students may subscribe to premium app services (~$1–$3/month), while universities may pay for station installation and analytics (~$2,000 per station initially).
- **Frequency of purchases:** Students pay monthly subscriptions; universities and advertisers are medium-term clients with semi-annual or annual contracts.
- **Trends:** Growing digital adoption among students and increasing reliance on public transport suggest strong, expanding demand for smart transport solutions.

3.1.5. Competitive intensity study in the industry



Customers/Buyers:

1. Students' price sensitivity and reliance on affordable transport: If prices are high or accuracy is low, students will easily switch back to normal buses or services. Strong Force
2. Universities as main institutional clients: Universities can negotiate prices or features, and their adoption decisions directly affect revenue. Strong Force
3. Tech-savvy users with high expectations : Students expect accurate timing and smooth app performance; bad experience quickly lowers trust. Strong Force

New Entrants:

1. Low entry barriers for tech startups : Other small companies could easily develop similar apps or display systems if they see your idea's success. Moderate Force
2. Need for partnerships and permits : New competitors might struggle to get municipal approvals or build local relationships as your company already did. Weak to Moderate Force
3. High setup and maintenance costs for physical stations : Building smart stations with LED screens and AI systems requires investment, which discourages many new entrants. Weak Force

Substitutes or Substitution:

1. Traditional buses and sérvices (shared taxis) : Students can still rely on regular transportation without using the smart station, making substitution easy. Strong Force
2. Ride-sharing and private apps (like Bolt or Careem) : Alternative transport apps already exist, offering convenience and flexibility that could replace your service. Strong Force
3. Student carpooling or university-organized transport : Some universities or students may organize their own transport systems, reducing dependence on the smart stations. Moderate to Strong Force

Public Power:

1. Government control over transportation and public area permits: The project depends on approvals from the Ministry of Transport and local municipalities, giving the public sector high influence. Strong Force
2. Municipal cooperation for station installation: Stations require space near universities or public roads, making municipal support essential for success. Strong Force
3. National push for smart infrastructure: The government's interest in smart city projects can help or delay your system depending on policy direction. Strong Force

Suppliers:

1. Hardware Providers (LED/LCD Screens):
The project depends on LED/LCD screen suppliers for displaying real-time data and ads. Few offer durable, outdoor-quality screens, but they're replaceable — making their influence moderate. - Moderate Force
2. Internet and Electricity Providers:
Reliable power and internet are vital for operation. Limited options in some Bekaa areas give these suppliers strong influence over system reliability.- Strong Force
3. Software and GPS Providers:
GPS data is collected from drivers' smartphones instead of external devices, reducing dependency. Software and cloud services are flexible and easy to switch - Weak Force

Competitive Rivalry:

1. Limited Local Competition:
The Smart Transport Hub operates in a new market with few existing smart transport systems, giving it a weak competitive force. - Weak Force
2. Emerging Market Potential:
As digital mobility grows, new competitors may appear, but strong partnerships with universities and transport companies can keep rivalry moderate.- Moderate Force
3. Unique and Innovative Solution:
Real-time tracking, AI predictions, and ad displays make the project stand out, keeping competition weak for now - Weak Force

Each force of competition can be countered by a series of elements, which constitute as many KSF when this force is preponderant.

Force of competition	Elements to counteract it
Threat of substitutes	• Offer unique value: real-time tracking + physical smart stations (not available in local apps like Bolt or Careem). • Use AI for accurate ETAs, which traditional transport lacks. • Build partnerships with universities to integrate service into campus life
Threat of potential entrants	• Establish strong early partnerships with universities and transport operators. • Secure municipal permits and approvals early to create barriers. • Build brand loyalty through reliable service and student engagement.
Bargaining power of buyers	• Offer tiered pricing: free basic info + premium features (e.g., seat booking). • Use advertising revenue to keep student subscriptions affordable. • Provide high-quality, reliable service to reduce willingness to switch
Bargaining power of suppliers	• Use driver smartphones for GPS to reduce dependency on hardware suppliers. • Partner with multiple LED screen suppliers to avoid monopoly dependence. • Explore renewable energy options (e.g., solar) to reduce reliance on grid electricity.
Role of public authorities	• Align with national smart city and transport initiatives to gain support. • Engage early with municipalities for permits and public space usage. • Ensure full compliance with data privacy and transport regulations

3.2 – Internal study and decission making:

3.2.1. SWOT analysis

Environmental scanning can influence an organization's strategy by creating both threats and opportuniti

Analysis of internal and external factors of the startup	
Strengths (Internal Factors)	Opportunities (External Factors)
• Real-time bus arrival info displayed on smart LED stations • AI-powered prediction system for accurate estimated times of arrival • Integrated platform connecting students, drivers, and transport companies • Affordable setup and scalable system • Localized solution specifically designed for Bekaa region conditions	• Growing adoption of digital and smart solutions among students • Government smart city initiatives and infrastructure support • Partnerships with universities and transport operators • Revenue potential from advertising, subscriptions, and sponsorships • Expansion potential to other regions or cities in Lebanon
Weaknesses (Internal factors)	Threats (External Factors)
• Initial development cost ($7,000–$10,000 for prototype) • Reliance on electricity and internet availability • Need for minor permits and approvals from local authorities • Dependence on driver cooperation for GPS tracking • Limited brand recognition in early stage	• Political instability and security concerns in Lebanon • Currency devaluation & inflation increasing hardware/software costs • Resistance from traditional drivers or transport companies • Data privacy regulations and GPS tracking compliance • Competition from alternative transport apps (Bolt, Careem

3.2.2- TWOS Matrix :

The TWOS matrix is built from the SWOT analysis conclusions or it allows to generate strategic options that correspond to different combinations of internal and external factors.

The TWOS matrix			
	Opportunity points		**Threat points**
Strengths Points	S / O: Use forces to seize opportunities:		S /T : Use forces to avoid threats:
	• Accurate AI + GPS system mitigates risk from unreliable traditional transport • .Affordable, scalable system helps cope with currency devaluation & inflation • Strong integration reduces dependency on single drivers, limiting resistance from traditional transport		• Leverage AI prediction and smart stations to partner with universities for pilot programs • Promote integrated platform & real-time info to attract advertisers and sponsors. • Expand the system to other regions based on initial local success
W ea	W / O: Minimize weaknesses to seize opportunities:		W /T : Minimize weaknesses to avoid threats:

	• Develop backup power and offline modes to reduce dependence on electricity/internet. • Plan phased rollout to manage initial costs and inflation risks • Build data privacy compliance into the app to avoid regulatory issues • Promote brand and local trust to overcome resistance from traditional drivers		• Collaborate with universities and municipalities to simplify permits and approvals. • Apply for government smart-city programs to fund station setup • Train drivers to reduce dependence on technical knowledge and improve cooperation

Startup execution stages

4.1- Conception Stage

4.1.1. The "Business Model Canvas"

Key partners	***Key activities***	***Offer (value proposition)***	***Customer relationship***	***Customer segments***
• Universities and colleges in the Bekaa • Public and private transport companies • Local municipalities • Technology suppliers (LED screens, GPS devices, software tools) • Advertising partners (local businesses and sponsors) **Resources & Roles:** • GPS and route data from transport companies • Station locations and electricity from universities and municipalities	**For Value Proposition:** • Design/build smart stations with LED screens and sensors • Develop/maintain AI for arrival predictions • Update mobile app for all users • Provide technical support and maintain data accuracy • Manage digital advertising content **For Distribution Channels:** • Integrate GPS data with AI system	• Smart, real-time, efficient transportation for students • Reduces waiting times and overcrowding • Smart stations with LED screens for live updates • Mobile app with GPS tracking and QR-based seat reservation • Real-time alerts for delays and route changes • Addresses punctuality, reliability, comfort, and safety needs	**Universities & Colleges:** • Long-term collaborative partnership • Co-development agreements for stations and secure data sharing • Moderate cost (coordination, installation, maintenance) **Students (End Users):** • User-focused, based on trust and real-time accuracy • Mobile app and smart station interface for live	• Students: real-time updates, reduced waiting times • Universities & Colleges: improved transport, safety, student satisfaction • Transport Companies & Drivers: GPS tracking, optimized routes • Advertisers & Local Businesses: platform to reach students **Primary Customers:** Students and universities **Secondary Customers:**

• Advertising content from local businesses • Hardware and installation from technology suppliers • Transport companies: GPS tracking and data sharing • Universities: provide locations, promote usage • Municipalities: permits and infrastructure support • Tech partners: supply, install, maintain equipment • Advertisers: provide content for revenue **Motivations:** • Reduce costs and risks • Access resources (data, hardware, physical space)	• Coordinate station installation and maintenance • Optimize app access to schedules/notifications • Promote and expand network to new universities/areas **For Customer Relationships:** • Responsive customer support • Gather/analyze feedback • Transparent communication with universities and transport providers • Offer personalized features (alerts, booking options) **For Revenue Streams:** • Manage ad contracts/schedules		updates and bookings • Low cost (app maintenance, support, minor marketing) **Transport Companies & Drivers:** • Operational and data-sharing partnership • GPS tracking and route data sharing • Moderate cost (integration, training, communication) **Advertisers / Local Businesses:** • B2B partnership for digital ads on stations • Advertising contracts established • Low cost, minimal client maintenance	Transport companies and advertisers **Model:** Multi-sided platform connecting students, universities, transport providers, and advertisers

• Increase market adoption	• Develop subscription systems • Maintain agreements with universities • Expand partnerships with transport providers and advertisers			
	Key resources • **Physical:** Smart stations, LED screens, GPS devices, servers, network infrastructure • **Intellectual:** Brand, software platform, AI algorithms, data models • **Human:** Developers, data engineers, technicians, marketing/support staff • **Financial:** Seed funding, operational budget		***Distribution channels*** • Universities, mobile apps, social media platforms • Early-stage: pilot projects with universities, online marketing • Integrated: app linked to smart stations via AI and GPS system • Most effective: university partnerships and mobile app engagement	

<table>
<tr>
<td colspan="2"></td>
<td colspan="2">Resource Requirements:
Smart station hardware and connectivity
AI prediction system
Mobile and web apps
Data servers/cloud storage
Customer support systems, analytics tools
Ad management software and payment systems</td>
<td colspan="2"></td>
<td colspan="2">• Most cost-efficient: digital and social media marketing

• Integrated into routines: students check bus times or book seats via phone</td>
<td colspan="2"></td>
</tr>
<tr>
<td colspan="5">Cost structure

• Major Costs: LED screens and installation, software/app development, cloud hosting, staff salaries, marketing, operations (electricity, internet, maintenance)

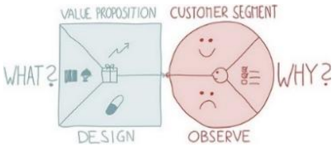
• Most Expensive Resources: LED screens, software development, cloud infrastructure, technical staff

• Most Expensive Activities: System/app development, hardware installation/maintenance, marketing and partnership management</td>
<td colspan="5">Sources of income
Customers willing to pay for: real-time updates, reliable info, premium booking
Current payment: basic transport, mostly cash
Preferred payment: digital subscriptions, university accounts, mobile payments
Revenue sources:
Smart station sales/leasing (main)
Advertising on LED screens (secondary, continuous)
Student subscriptions and transport partnerships (recurring)
Optional government/municipal support</td>
</tr>
</table>

• **Business Type:** Value-driven (smart, efficient transport) • **Cost Characteristics:** Fixed (staff, hosting, app maintenance), variable (electricity, internet, marketing), economies of scale as stations expand	

4.1.2. The value proposition canvas

It represents the group of products and services that creates value for the customer. It aims to address an unmet customer need and improve its current situation.



Collaborators side

Collaborators :

Universities: Provide access to campus transportation data and allow station installations.
Transport Companies & Drivers: Enable GPS tracking and data sharing.
Advertisers / Local Businesses: Use the platform for ad placements.
Tech Developers & Engineers: Maintain AI, mobile app, and hardware systems.

Developing and maintaining the AI prediction system.

Managing and updating the mobile app for students and drivers.

Installing and servicing smart stations.

Handling partnerships, advertisements, and customer support.

Expanding to new universities and regions.

Tasks

Pains

Initial cost and technical setup challenges

Training drivers and users on the system

Maintenance and connectivity issues

Data privacy and system reliability concerns

Universities: organized transport and safety for students

Transport companies: efficient routes and satisfied clients

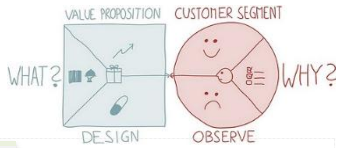
Drivers: clear schedules and less stress

Advertisers: targeted local exposure

Developers: innovative project and tech growth opportunity

Desired gains

Value Proposition Side (Product / Service)



Smart Stations (Hardware): Equipped with LED screens showing real-time transport information, ads, and alerts.

Mobile Application (Software): For students to view arrival times, book seats, and receive delay notifications.

Driver / Transport App: Used by drivers or transport companies to share live GPS data.

AI Data System: Analyzes GPS and route data to predict accurate arrival times.

Advertising Platform: Allows local businesses to display digital ads on station

Characteristics of the products or services

Pain Relief

Eliminates waiting uncertainty caused by late or unpredictable transport.

Reduces crowding and confusion at bus stops.

Solves the problem of lack of coordination between drivers and students.

Minimizes communication gaps by providing instant notifications and real-time updates.

Offers an affordable and scalable solution for universities and companies.

Increases time efficiency for students by reducing uncertainty in transport schedules.

Enhances safety and comfort by providing organized waiting areas.

Helps universities improve their transportation management and student satisfaction.

Gives transport companies better route monitoring and reliability.

Generates extra revenue through advertising and subscriptions.

Gains Creator

4.2- Quality Management System Study

PDCA (Plan-Do-Check-Act) method, also known as the Deming Wheel, forms the core of the structured quality management framework adopted to ensure the reliability, usability, and continuous improvement of the Smart Transport Hub platform. Aligned with international quality standards and best practices in tech-driven mobility solutions, PDCA guides development from prototype to full-scale deployment by integrating systematic planning, iterative testing, performance evaluation, and responsive refinement. This approach ensures the delivery of a high-quality, user-centered smart transport system that meets the real-time information needs of students, drivers, and transport operators in the Bekaa region—while remaining scalable, sustainable, and adaptable to feedback and changing market conditions.

- The PDCA cycle has been specifically adapted to:
- **Better understand the needs of applicants** (students, drivers, universities, and advertisers).
- **Implement the processes necessary to achieve these needs** through systematic development and deployment.
- **Measure the satisfaction of these applicants and evaluate the performance of the processes implemented**.
- **Make improvements and correction decisions** to continuously enhance platform processes.

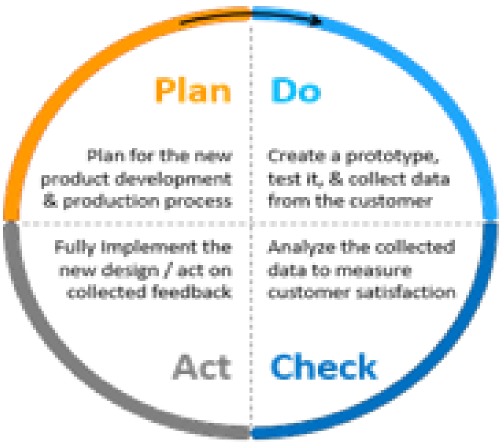


Table: Detailed PDCA Activities for Smart Transport Hub

Phase	Key Activities	Objectives / Outputs
PLAN	• Conduct surveys & interviews with students, drivers, and universities.	• Clear user requirements document. • MVP feature list & development roadmap.
	• Define MVP features: smart LED station, driver app, AI prediction.	• Measurable quality targets (KPIs).
	• Set KPIs: ETA accuracy >90%, station uptime >95%, user satisfaction >4/5.	• Selected pilot location: University main entrance.
	• Select pilot location: University main entrance.	
DO	• Install solar-powered LED station with real-time display.	• Functional prototype at pilot location. • Driver app deployed and in use.
	• Develop driver app with GPS activation & route logging.	• AI model generating live ETAs.
	• Train 10 pilot drivers on app usage.	
	• Run AI prediction model with historical & live GPS data.	
CHECK	• Collect feedback via in-app forms & QR codes at station.	• Feedback report & satisfaction scores. • Performance dashboard with KPIs.

	• Monitor system logs: uptime, prediction delay, GPS accuracy.	• Identified areas for improvement.
	• Compare predicted vs. actual arrival times.	
	• Conduct focus groups with students & drivers.	
ACT	• Update AI model with real traffic patterns.	• Improved system accuracy & usability. • Scalability plan for further deployment.
	• Redesign app UI/UX based on usability feedback.	• Documented SOPs for ongoing quality management.
	• Expand to 2 new stations in high-demand areas.	
	• Formalize maintenance and support procedures.	

4.2.1. Main Processes description

Process Title	Process Actors	Process Description	Process Documents/Resources	Remarks
Driver Registration and GPS Activation	Driver – Transport Company – AI System	The driver downloads the **Smart Transport Hub** mobile application and creates an account by entering basic details such as name, vehicle number, and assigned route. After registration, the driver activates GPS tracking through the app, which automatically shares real-time location data with the system. The AI platform monitors the vehicle's movement, calculates estimated arrival times to each smart station, and can predict arrival times even during temporary connection loss using the last known location.	Mobile application – Driver's smartphone GPS – Internet or mobile data connection – Database – Server.	The system requires an active internet or mobile data connection for live tracking. If the driver temporarily loses connection, the app stores the last known location, and the AI system estimates arrival times using predictive algorithms. Tracking resumes fully once the connection is restored. GPS runs directly from the driver's phone, so no external hardware is needed.

Smart Station Display Update	AI System – LED Display – Server	The AI system continuously receives real-time location data from drivers' GPS. It processes the data and updates the LED screen at each smart station with accurate arrival times, vehicle numbers, and route information. The update happens automatically every few seconds to ensure real-time accuracy.	LED display screen – Central server – Internet connection – GPS data from drivers – AI software	The system works automatically without manual input. Internet stability is required for continuous updates.
Passenger Information and Notification Process	Student (Passenger) – Mobile Application – AI System	Passengers (students) access the mobile app to view real-time vehicle status, estimated arrival times, and available seats. They can also reserve a seat using a QR code. The system sends automatic notifications about any delays, cancellations, or changes in arrival times.	Mobile application – Smartphone – Internet connection – Notification system – QR code generator	Notifications are automated and synchronized with the AI prediction system to ensure accurate information delivery.
Advertisement and Technical Maintenance Process	Advertising Partner – Technical Team – LED Screen System	Local businesses or advertisers upload their ads through an online management platform. The ads are scheduled and displayed on the LED screens at smart stations. Meanwhile, the technical team monitors station performance, performs maintenance, and ensures proper system updates.	Advertisement management platform – LED display system – Maintenance tools – Internet connection	Regular maintenance is required to ensure continuous operation and high display quality. Ad content is moderated before publishing.

4.2.2 Evaluation tools

Quality objectives have been set with performance indicators for each of these objectives, the purpose of which is to compare them with the measured results.

Table of objectives

Goal	KPI	Formula	Target	Frequency of measurement
1. Maximize student satisfaction	Average satisfaction score	Sum of all satisfaction ratings ÷ total responses	**≥ 4/5 average score**	*End of each quarter*
2. Improve arrival-time accuracy	ETA prediction accuracy rate	(Number of accurate ETAs ÷ total ETAs) × 100 Monthly	**≥ 90% accuracy**	*At the end of each quarter*
3. Resolve all reported issues quickly	% of issues resolved within 24–48 hours	(Issues resolved within 48 hr ÷ total issues) × 100	**100%**	*At the end of each quarter*
4. Maximize risk prevention	Number of prevention actions implemented	Total number of risk-prevention actions per year	**10–15 actions/year**	*At the end of each quarter*
5. Respect the project budget	Budget efficiency rate	(Budget spent ÷ total allocated budget)	≤ 1	*At the end of each quarter*

6. Optimization of the 31xécution of the marketing plan	% of marketing actions completed	(Completed marketing actions ÷ planned actions) × 100	100%	*At the end of each quarter*

4.2.3. Risk assessment and prevention

The risks	Potentiality	Impact	The causes	Potential impacts	Preventive action
Platform or App Unavailability	Medium	High	Internet outage at a location - Server failure or maintenance - DoS attacks - App bugs or crashes	Students cannot access real-time bus info – Missed notifications about delays Loss of trust in system – Reduced adoption by students and universities	– Implement an offline mode in the app that caches schedules and last known data. – Use redundant servers and backups to avoid downtime. – Schedule maintenance during low-usage hours. – Apply security measures to protect against DoS attacks and app vulnerabilities. – Monitor app and server status in real-time to detect and fix issues quickly
GPS Tracking Failure	Medium	High	GPS signal weak in certain areas - Driver forgets to activate GPS - Driver's phone battery dies	- AI cannot calculate accurate arrival times - Students wait unnecessarily - Frustration and loss of trust in the system - Bad reputation for the platform	- Notify drivers to activate GPS at start of route - Use AI prediction based on last known location during temporary offline periods - Send alerts if driver's device is offline or battery is low

			- Phone offline / no mobile data		- Provide training for drivers to ensure correct app usage
Incorrect Arrival Time Display	Medium	High	AI prediction errors, outdated traffic data, unexpected delays such as accidents or roadblocks.	Students may arrive late or miss transport; decreased trust in the system's accuracy.	Continuously update AI algorithms with real-time traffic data; integrate multiple data sources (GPS, driver input); allow manual driver updates in case of incidents.
Delayed Communication Between Drivers and System	High	Medium	Driver forgets to start the app, weak or lost mobile data connection, delayed server response.	Arrival times not updated; students misinformed; reduced system reliability.	Add automatic GPS activation when vehicle starts; send app reminders to drivers; use local data caching to update when connection is restored.
Hardware Failure at Smart Station	Medium	High	LED malfunction, power outage, or vandalism.	No display of arrival times or ads; confusion for users; loss of ad revenue.	Regular hardware maintenance; use surge protection; install CCTV or physical protection; keep backup screens available.
Data Privacy and Security	Medium	High	Unauthorized access to user or GPS data, app vulnerabilities, weak encryption.	Breach of privacy, legal issues, loss of user and university trust.	Apply strong encryption and secure servers; comply with local data protection laws; conduct regular security audits; restrict admin access.

Customer Refusal / Complaints	Medium	Medium	Dissatisfaction with accuracy, poor performance, or undelivered ads.	Negative feedback, reduced adoption, and potential revenue loss.	Collect user feedback regularly; improve system accuracy; provide technical support and compensation for missed ads.
Business / Financial Risks	High	High	Low adoption rates, insufficient ad revenue, high maintenance cost.	Reduced profitability; slow project growth; possible shutdown.	Diversify income sources; offer subscription models; seek university partnerships and government support; manage costs carefully.
Natural or Environmental Risks	Medium	Medium	Heavy rain, extreme heat, floods, or power cuts.	Station downtime; damage to screens or systems; interruption of service.	Use waterproof and heat-resistant materials; install backup power systems; perform regular environmental protection checks.

4.3. Marketing Plan :

4.3.1. Strategic Marketing (segmentation,targeting and positioning)

Customer Target Segmentation :

The adapted segmentation variables differ for each of the collaborators:

- **Collaborator 1: Students**
  - Location: Bekaa-based students
  - Frequency of transportation use: daily commuters
  - Need for accurate timing & notifications
  - Smartphone usage and digital readiness
- **Collaborator 2: Drivers & Transport Companies**
  - Type of transport: van, bus
  - Willingness to use mobile GPS tracking
  - Route consistency: regular vs. irregular
  - Digital adoption level
- **Collaborator 3: Advertisers & Local Businesses**
  - Type of business: student-targeted or general
  - Advertising budget
  - Need for digital visibility
  - Interest in promoting on university routes

Targeting :

Targeting variables adapted for:

- **Collaborator 1: Students**
  - Segment selected: Daily commuting students relying on buses/services
  - Operational choice: Undifferentiated marketing (one product for all students)
- **Collaborator 2: Drivers & Transport Companies**
  - Segment selected: Professional transport companies and drivers with fixed routes ready to activate GPS via mobile app
  - Operational choice: Differentiated marketing (training + support for companies)
- **Collaborator 3: Advertisers & Local Businesses**
  - Segment selected: Small local businesses with low budgets targeting students
  - Operational choice: Undifferentiated marketing (same ad format for all advertisers)

The Positioning :

To position the **Smart Transport Hub** effectively, a competitor analysis was conducted to understand the market environment. The main competitors include **traditional public transportation services**(buses, vans, and service cars without digital systems) and **basic transportation apps** (e.g., Bolt or AlloTaxi) that provide limited tracking or communication features.

Traditional transport providers offer low-cost services but lack real-time tracking, structured scheduling, and reliability. Their weaknesses include inconsistent timing, no communication strategy, and low customer service. Their reputation varies and depends mostly on individual drivers rather than an organized system.

Basic transport apps offer simple location-sharing or booking features but do not provide smart stations, AI arrival prediction, or LED display systems. These apps usually rely on user-generated information, which limits accuracy and consistency. Pricing varies but is generally affordable, and their communication is mostly digital.

In comparison, the Smart Transport Hub provides **real-time GPS tracking**, **AI-based arrivalprediction**, **LED station displays**, and **integrated driver applications**, offering higher service qualityand reliability.
The project positions itself as a **modern, accurate, and student-focused solution** thatbridges the gap between traditional transport and smart mobility systems.

Importance to customers

1 = most important, 5 = not important

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	Point over 5	Strength points	Weakness points	Point over 5	Strength points	Weakness points	Point over 5	Strength points	Weaknes s points
Product/Services	Real-time GPS tracking from driver phones AI-based arrival prediction LED/LCD station screens for live updates Delay notifications & route information	Requires internet connection for stations and drivers Hardware installation cost (LED/LCD screens) Depends on drivers	4	Already widespread Simple operation Low cost	No real-time tracking No fixed or reliable arrival times Zero communication with users No digital services or data	2	GPS tracking for ride arrival Digital booking Easy communication with driver	Not designed for buses or university transport No stations or LED screens No AI predictions Not integrated with student communities	3	1	Improve stability of GPS tracking Add seat reservation or student-only features Increase station coverage

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	**Point over 5**	Strength points	Weakness points	Point over 5	**Strength points**	Weakness points	**Point over 5**	Strength points	Weaknesses points
	Advertising display for extra revenue	activating the app									around universities Enhance AI accuracy with more data
Price	Affordable for universities (rent or purchase model) Additional revenue through advertising reduces overall cost Free basic use for students	Initial installation cost for LED/LCD stations Ongoing internet + maintenance fees	3	Very low cost to operate No digital infrastructure needed Drivers set cheap, familiar prices	No added value for the price (no tracking, no system, no reliability) Unorganized pricing (varies by driver, inconsistent)	2	Clear pricing inside the app Promotions and discounts sometimes available	More expensive than buses/services Not affordable for daily student transport Pricing varies based on demand	4	2	Offer flexible subscription plans for universities Provide student premium features at low cost Use advertisements to reduce overall station pricing Partner with universities to share

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	**Point over 5**	Strength points	Weakness points	Point over 5	**Strength points**	Weakness points	**Point over 5**	Strength points	Weakness points
											installation cost
Quality	Accurate real-time GPS tracking AI-generated arrival predictions (high precision) LED/LCD screens with clear visibility Reliable digital information (less human error)	Quality depends on internet stability AI accuracy improves only with more data over time Possible hardware issues (screen damage, weather impact)	4	Simple system (no technology to break) Experienced drivers	Very inconsistent timing No tracking → very low reliability Unpredictable routes or delays No information quality for customers at all	1	Good app reliability Accurate driver tracking Real-time notifications	Not suitable for buses or group transport No station displays No AI prediction Different quality depending on driver phone/internet	3	1	Not suitable for buses or group transport No station displays No AI prediction Different quality depending on driver phone/internet
Selection	Easy for students to use — just check the LED screen Drivers only need a mobile app (no special hardware) Stations placed in main student areas	System activation depends on drivers turning on the app Limited station coverage at launch	4	Very easy to access: buses/services pass frequently in main roads No app or technology required	Unpredictable — students don't know which bus is coming or when No organized selection → student must wait long times	2	Easy app booking Clear driver selection and pickup location	Not designed for student stations Not available in many Bekaa regions	3	1	Increase station locations around universities Encourage drivers with rewards to keep app activated

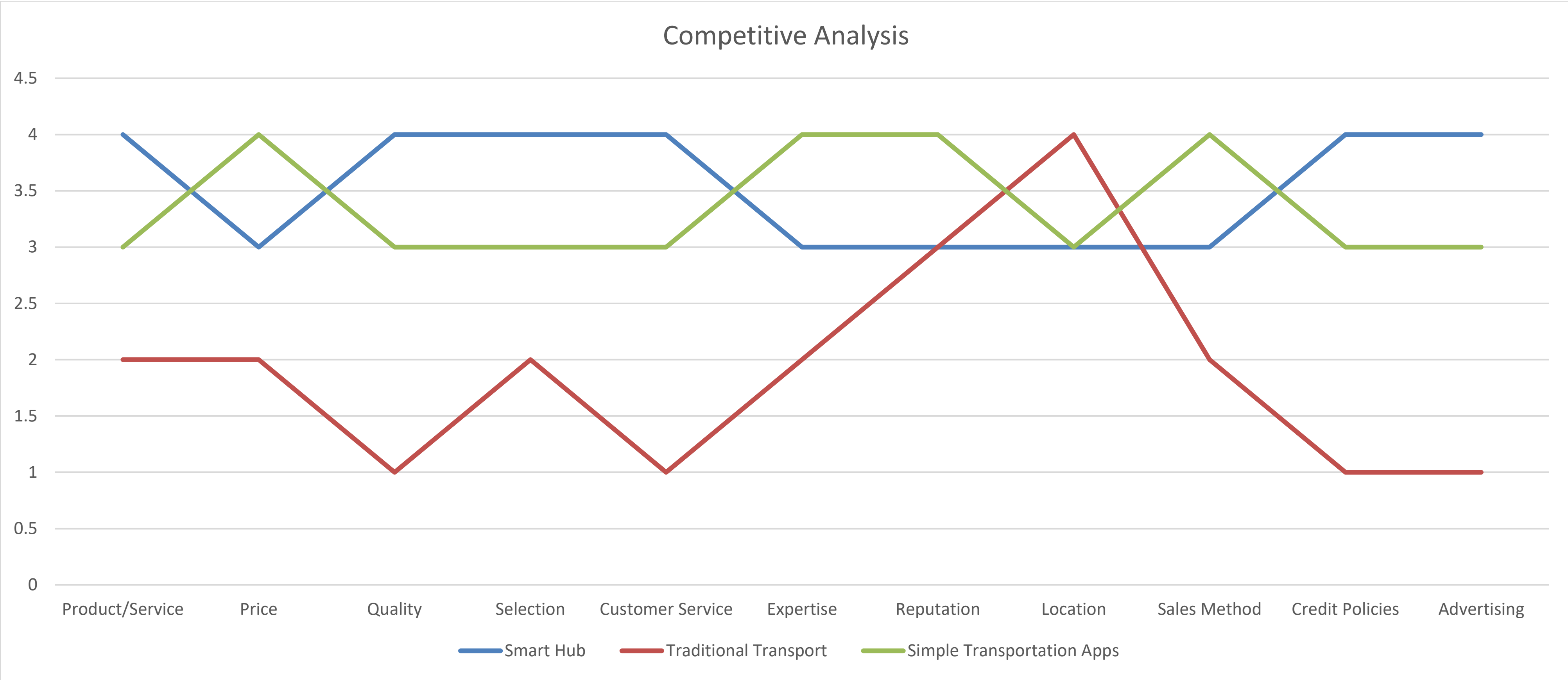
Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	**Point over 5**	Strength points	Weakness points	Point over 5	**Strength points**	Weakness points	**Point over 5**	Strength points	Weakness points
											Add student ID login for prioritization
Customer Service or after sale service	Digital support (app, website) Real-time problem reporting Automatic alerts and notifications	Requires internet + support team New startup → reputation still building	4	Drivers sometimes flexible / personal relationships	No customer service at all No complaint system No organized management	1	In-app support and reporting Driver ratings → improve service	Not designed for buses or university transport Limited support in local areas	3	3	Build a quick-response support team Add QR code at stations for instant reports Build trust through campus partnerships
Expertise	Specialized in smart mobility, GPS tracking, AI prediction, and digital displays. Tech-driven system.	New startup with limited field experience at launch.	3	Long experience in routes and driving.	No technology expertise, no data systems.	2	High software expertise and experience in ride-tracking.	Not designed for bus systems or stations.	4	3	Improve staff training in AI, data analytics, and user support. Build partnerships with universities to gain

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	**Point over 5**	Strength points	Weakness points	Point over 5	**Strength points**	Weakness points	**Point over 5**	Strength points	Weaknesses points
											credibility and expertise.
Company Reputation	Innovative, modern, student-centered, professional branding.	New in the market; reputation still forming	3	Familiar & known by students for years.	Reputation depends on individual drivers; inconsistent.	3	Strong brand recognition (Bolt/Allo Taxi).	Not known in the bus/service sector; limited local presence.	4	2	Build trust with pilot stations, student testimonials, social proof, and university endorsements. Run awareness campaigns, offer demo stations, provide guaranteed service quality.
Location	Stations placed in high-traffic university areas; easy visibility.	Limited coverage in the first phase.	3	Available across cities; everywhere on the street.	No organized station locations.	4	Available wherever smartphones work.	Not available in remote Bekaa areas due to lack of drivers.	3	2	Expand stations to more campuses, high-density student areas, and main roads.

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness points	Point over 5	Strength points	Weakness points	Point over 5	Strength points	Weakness points	Point over 5	Strength points	Weakness points
											Work with municipalities & universities to secure important station spots.
Sales Method	Direct sales to universities; subscription model; advertising partnerships.	Requires formal agreements with institutions.	3	Direct cash payment; simple.	No system for organized sales; inconsistent fares.	2	Online booking; digital payments.	Not adapted for buses / group transport deals.	4	3	Offer flexible packages (monthly /annual), add pay-per-use options for drivers. Use demos and pilot testing to simplify sales conversations with universities.
Credit Policies	Can offer installment or rental options to universities; ad revenue lowers cost.	Requires financial planning for hardware installations.	4	Pay-as-you-go only.	No credit options.	1	Secure in-app payments; digital receipts.	No credit for service vans or buses.	3	4	Provide flexible financing plans; allow shared cost models

Factor	Smart Transport Hub			Traditional Transport			Simple transportation apps			Importance to Customer	Action to work on for improving
	Strength points	Weakness **points**	**Point over 5**	Strength **points**	Weakness **points**	Point over 5	**Strength points**	Weakness **points**	**Point over 5**	Stren gth **points**	Weaknes s **points**
											with partners. Offer sponsorships or co-branding to help cover station costs.
Advertising	LED screens allow digital ads: revenue stream for university & company.	Requires partnerships with advertisers; needs good traffic to attract sponsors.	4	Word-of-mouth only.	No formal advertising channels.	1	Online digital ads inside apps.	App ads not visible to students waiting at stops.	3	3	Sell ad slots to local businesses; create student-targeted ad packages. Improve screen design, add rotating ads, track ad views to attract clients.

Competitive Analysis



4.3.2. Operational Marketing (4 P of the marketing mix)

The marketing mix:

"The marketing mix (sometimes referred to as a market plan or a marketing policy) refers to the coherent set of product, price, distribution and communication decisions of a company's or brand's products.

Product / Service Policy:

Major Products/Services of Smart Transport Hub:

1. **Smart Stations (Hardware)**
    - **What it is:** Physical bus/van stops equipped with LED screens displaying real-time arrival and route information.
    - **How it meets customer needs:** Provides commuters with accurate, real-time information, reducing waiting time, uncertainty, and overcrowding.
    - **Key Features / Special Aspects:** LED screens, connectivity to GPS-enabled driver apps, durable for outdoor conditions.
    - **Benefits to Customers:** Students and commuters can plan their trips efficiently, avoid delays, and gain a safer, more organized waiting environment. Advertisers gain visibility on high-traffic screens.
2. **Mobile & Web Application (Software for Students & Commuters)**
    - **What it is:** A digital platform for tracking vehicles, reserving seats, and receiving alerts.
    - **How it meets customer needs:** Helps students and commuters save time, book trips in advance, and receive instant notifications about delays or schedule changes.
    - **Key Features / Special Aspects:** GPS-based vehicle tracking, QR-code seat reservation, push notifications, user-friendly interface.
    - **Benefits to Customers:** Reduces uncertainty, improves punctuality, and enhances convenience in daily transport management.
3. **Driver / Transport Operator App (Software for Transport Companies)**
    - **What it is:** An app used by drivers to share live GPS data and update routes in real-time.
    - **How it meets customer needs:** Ensures accurate arrival predictions and smooth coordination between transport operators and commuters.
    - **Key Features / Special Aspects:** Easy GPS integration, route management, real-time status updates.
    - **Benefits to Customers:** Reduces operational stress for drivers, improves reliability, and helps transport companies monitor and optimize routes efficiently.
4. **AI-Based Arrival Prediction System (Software Service)**
    - **What it is:** An artificial intelligence platform that predicts bus/van arrival times based on GPS, speed, traffic, and historical data.

- **How it meets customer needs:** Provides highly accurate arrival times, reducing waiting time and improving planning for commuters.
- **Key Features / Special Aspects:** Predictive algorithms, continuous learning from traffic patterns, integrates with stations and mobile apps.
- **Benefits to Customers:** Commuters get reliable information, universities can improve transport planning, and operators optimize schedules.

5. **Advertising Platform (Service for Businesses)**
   - **What it is:** A system that allows businesses to display digital advertisements on smart station screens and the mobile app.
   - **How it meets customer needs:** Offers local businesses a targeted platform to reach students and commuters efficiently.
   - **Key Features / Special Aspects:** Digital ads on LED stations, app integration, flexible pricing for advertisers.
   - **Benefits to Customers:** Businesses increase visibility and engagement; the system generates additional revenue to support operational costs.

Name, logo and slogan of the product / service:

Product name justification :

rAsif

Justification : **rAsif** is a unique and memorable brand name derived from the initials of the founding team members (Rana, Sarah, Israa, and Fatima), highlighting the collaborative identity behind the project. Beyond its symbolic origin, the name has a meaningful linguistic relevance: in Arabic, **"رصيف / Raseef" means "sidewalk" or "part of the station,"** which directly connects to the startup's mission of enhancing transportation access and organization. This dual significance gives the brand both a personal and contextual identity — it is rooted in teamwork while also being naturally associated with mobility services.

The name is short, modern, and easy to pronounce in multiple languages, making it appropriate for a tech-driven solution aimed at youth. Its distinctiveness, along with its semantic relation to transport infrastructure, strengthens brand recognition and makes it memorable in a market where names are often generic or highly descriptive.

Slogan justification :

Slogans :

1) Know the When, Master the Day.

2) Smart Rides for Smart Minds.

3) On Time. Every time.

4) Never Wait Blindly.

Justification : We selected a set of slogans that communicate reliability, intelligence, and time-efficiency, which are the core values of the rAsif service. Each slogan focuses on solving a major pain point — unpredictable waiting time — in a simple, memorable, and motivational tone. Whether emphasizing punctuality ("On Time. Every Time."), awareness ("Know the When, Master the Day."), intelligence ("Smart Rides for Smart Minds."), or uncertainty reduction ("Never Wait Blindly."), the slogans reinforce the brand promise of a smarter, more organized transportation experience for students.

Logo justification :

Justification : The rAsif logo visually communicates the brand's identity as a modern, technology-driven transportation solution. The central road leading to an arrow symbolizes movement, progress, and the promise of efficient, on-time transportation. The bus icon represents the core service — organized and accessible public transport — while the rising arrow reflects improvement, direction, and smart planning throughout the journey.

The logo integrates network nodes and a wireless signal, highlighting the use of GPS, connectivity, and digital intelligence in managing transport operations. This conveys that rAsif is not just a mobility service, but a connected ecosystem that uses data to enhance user experience.

The blue color palette reinforces values such as trust, reliability, technology, and safety, which are essential for transportation services used daily by students. The modern, rounded typography paired with the tagline *"The Smart Transport Hub"* reinforces rAsif's positioning as an innovative, student-focused, tech-enabled platform.

Overall, the logo effectively communicates the brand's mission: smart, connected, and trustworthy mobility.

Promotion

The startup will promote its service using a digital-first strategy focused on social media to reach students effectively. The core message is: *"Smart, reliable, and affordable transportation for daily commuters."* Promotion will rely mainly on Facebook and Instagram ads, run monthly, because they offer low-cost targeting and high engagement. Additional tactics include flyers, QR codes on campus, and WhatsApp groups to encourage word-of-mouth. These methods were chosen because they are affordable, localized, and widely used by the target audience, making them effective for fast awareness and adoption.

	Year 1 /$		*Year 2 /$*		*Year 3 /$*		*Year 4 /$*		*Year 5 /$*	
	Details	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*
Emails Campaign X Dedicated IP X Template (design) Advanced Reporting Monitor (Who Clicked & Viewed) X Shots Responsive Design Occupation Specific	Launch announcement emails to all Bekaa students, university staff, and transport companies. 4 dedicated campaigns, advanced reporting on clicks/views.	$500	Follow-up campaigns highlighting app updates, new stations, and special features. 3 campaigns.	$300	Monthly newsletters with real-time updates, new ads, and AI features. 4 campaigns.	$300	Personalized emails for premium features and seat reservation promotions. 4 campaigns.	$300	Annual review campaigns showing system impact and benefits to stakeholders. 3 campaigns.	$300
Facebook Campaign Cost Per Click (CPC)Click to our site/Facebook page Cost Per Mile (CPM) Impression of our ads (site or page)	Launch campaigns with engaging posts, student testimonials, and video demos. CPC & CPM ads targeting Bekaa students.	$700	Seasonal promotions and updates on new smart stations. Targeted retargeting campaigns.	$500	Ads for premium features and partner promotions. Instagram stories highlighting app usability.	$500	Highlight AI features and real-time arrival predictions. Engage university communities.	$500	Community-driven campaigns: contests, polls, and user-generated content to increase adoption.	$500

	Year 1 /$		Year 2 /$		Year 3 /$		Year 4 /$		Year 5 /$	
	Details	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*
Google Adwords *SEO (Search Engine Optimasation)*	Paid search campaigns for students searching for transport info. Optimize website for organic search with keywords like “Bekaa bus tracker”.	$800	Expand keyword targeting, focus on transport companies and advertisers. Optimize blog content.	$600	Launch retargeting campaigns for users who visited the website but didn’t download the app.	$600	Regional expansion keywords and SEO updates for additional universities.	$600	Optimize for mobile searches, maintain high-ranking content, promote app updates.	$600
SMS	Pilot SMS notifications to early adopters: 500–1000 messages/month.	$200	Regular alerts for delays, app updates, and promotions. 1000–2000 messages/month.	$150	Expand to more users as adoption grows, with targeted alerts.	$150	Focused notifications for premium app users.	$150	Seasonal and strategic promotional SMS campaigns.	$150

	Year 1 /$		Year 2 /$		Year 3 /$		Year 4 /$		Year 5 /$	
	Details	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*
Leaflet+Bussiness card	Distribution at all major campuses and bus stops during launch phase.	$300	Refresh leaflets with app updates, QR codes, and new station locations.	$200	Focused campaigns at new station areas and student gatherings.	$200	Highlight premium features and partnership options for advertisers.	$200	Rebranding/relaunch with user success stories.	$200
Events	Launch events at universities with live demos of smart stations and app. Include a small charity drive (donation boxes or volunteering campaigns) to build community goodwill.	$500	Participation in student fairs or tech events, showcase AI arrival prediction, and organize a charity mini-marathon or blood donation drive on campus.	$300	Organize workshops on app usage, seat reservation, and feedback sessions, combined with charity activities like collecting school supplies for underprivileged students.	$300	Host community engagement events highlighting efficiency and time-saving benefits, plus charity campaigns such as environmental cleanup or awareness campaigns.	$300	Anniversary events, celebrating adoption and growth, with integrated charity events such as free transport days for students in need or sponsorship of local NGOs.	$300

	Year 1 /$		*Year 2 /$*		*Year 3 /$*		*Year 4 /$*		*Year 5 /$*	
	Details	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*	*Details*	*Amount*
Blogger	Collaborate with 3–5 local student bloggers for reviews and demos.	$400	Engage micro-influencers to post tutorials and benefits.	$300	Expand influencer network to other Lebanese cities if pilot successful.	$300	Promote AI features and premium app functionality via influencers.	$300	Showcase success stories and user experiences on social media/blogs.	$300
Radio/TV	Short radio spots during peak commuting hours in Bekaa.	$600	TV spots on local channels targeting students & commuters.	$400	Repeat campaigns for new station openings or updates.	$400	Focused campaigns on premium app features and advertising potential.	$400	Highlight adoption success, reliability, and expansion.	$400
Total		$4000		$2750		$2750		$2750		$2750

Pricing

Pricing Strategy and Structure:

- **Smart Stations:** Universities and municipalities can rent or purchase stations. Subscription or leasing options available.
- **Mobile App:** Basic features free for students; premium features (seat reservations, alerts) at $1–$3/month.
- **Advertising Platform:** Paid per impression (CPM) or click (CPC) on LED stations and app.

Discount Structure:

- Early adopter discounts for universities and transport companies.
- Volume discount for multiple stations or longer-term subscriptions.

Payment Policy:

- Payment via bank transfer, credit card, or online payment portals.
- Flexible payment plans for universities or municipalities (quarterly or annual installments).

Place

Physical Locations: Smart Stations

- **Where:** Installed at universities, busy bus stops, and student hubs in the Bekaa region.
- **Advantages:**
  - High visibility to target audience (students and daily commuters).
  - Convenient access near campuses and main roads.
  - Enhances brand image as modern and tech-savvy.
- **Disadvantages:**
  - Requires electricity and internet.
  - Permits and coordination with municipalities/universities needed.
  - Limited coverage at initial rollout.
- **Importance to Customers:**
  - Essential — stations must be at natural student gathering points and high-traffic commuting areas.
  - Easy accessibility, clear signage, and safety are critical.
- **Surrounding Businesses & Access Routes:**
  - Locations chosen near cafeterias, libraries, or campus entrances for high foot traffic.
  - Easy vehicular access for buses and vans.
  - Parking availability nearby helps students who come by car or bike.
- **Competition Location Analysis:**
  - Competitors (basic bus stops, ride apps like Bolt/AlloTaxi) do not have smart stations.

    - Placing stations near competitors is not necessary; strategic visibility and accessibility matter more.

Virtual Locations: Mobile & Web App

- **Where:** Accessible globally via smartphones and web browsers.
- **Advantages:**
    - 24/7 access for students and drivers.
    - Scalable across regions without physical installation costs.
- **Disadvantages:**
    - Requires stable internet and device compatibility.
- **Customer Convenience:**
    - App allows students to check real-time arrivals, reserve seats, and get notifications from anywhere.
- **Integration:**
    - Seamlessly connected to smart stations, enabling a consistent user experience across physical and digital platforms.

Distribution Channels

- **Retail / Physical Channels**

  - Smart stations installed at universities, bus stops, and student hubs.
  - Students access information directly at the station.

- **Direct / Digital Channels**

  - Mobile app (iOS & Android) for real-time bus tracking, notifications, and seat reservations.
  - Web platform for desktop users and university administrators.

- **Wholesale / Bulk Partnerships**

  - Contracts with universities and municipalities to install multiple smart stations at once.
  - Partnering with transport companies to integrate GPS tracking for all vehicles.

- **Own Sales Force**

  - Internal team responsible for:
    - Negotiating contracts with universities, transport companies, and advertisers.
    - Maintaining stations and supporting technical operations.

- **Agents / Independent Representatives**

  - Local representatives to manage relations with drivers and small transport operators.
  - Ensure adoption and proper use of the driver mobile app.

- **Bid on Contracts**

- Opportunities to bid on government or municipal smart-city infrastructure projects.
- Potential expansion into other Lebanese regions through public tenders.

4.4 Financial study :

4.4.1. Cost Structure :

This section presents the **monthly cost structure** of the Smart Transport Hub project, divided into **fixed costs** and **variable costs**. It includes all essential expenses required to operate the smart stations, mobile applications, and AI tracking system, along with operational costs that fluctuate based on activity levels.

Monthly Cost Summary

Fixed Cost

Category	**Estimated Monthly Amount ($)**	**Description**
Administrative Fees	2 500	Business registration, legal documents, accounting, system licensing.
Permanent Staff Salaries	12 000	Core team: software engineer, mobile developer, operations manager, technical lead, admin support.
Cloud Hosting & Server Costs	1 000	GPS tracking server, AI prediction engine, database, dashboard hosting.
Office Rent & Utilities	3 500	Rent, electricity, water, internet, storage for station parts.
Customer Support Operations	1 500	Support staff or outsourced service for drivers/students/universities.
Marketing & Branding Retainer	800	Monthly branding, website updates, promotional materials, social media presence.

Category	Estimated Monthly Amount ($)	Description
Total Fixed Costs	**21 000**	**Total = 2 500+12 000+1 000+3 500 +1 500+800= 21 000**
Variable Cost		
Primes - Bonuses	300	Monthly incentives paid to drivers who keep the tracking app active.
Travel Costs (Voyages)	150	Maintenance visits + installation trips in Bekaa.
Materials (Matériaux)	100	Cables, connectors, installation parts, small repairs.
. Commissions (Commissions)	200	Paid to advertising partners or sales agents.
Production Supplies (Fournitures de production)	80	Cleaning supplies, SIM replacements, protective covers.
Production-Based Salaries (Salaires basés sur la production)	400	Technician payments based on number of stations serviced.
Total Monthly Variable Costs	**1 230**	**Total = 300 + 150 + 100 + 200 + 80 + 400**
Total Monthly Cost	**22,410**	**Total Fixed Cost + Total Variable Cost**

Conclusion

This cost structure helps the Smart Transport Hub understand the essential fixed expenses required to keep the smart stations, software, and tracking system operating smoothly. At the same time, it allows flexible management of variable costs, which change depending on driver activity, station maintenance needs, and the level of system usage.

Detailed financial data can be monitored and adjusted as the number of active drivers increases, more smart stations are installed, and new universities or partners join the platform. This ensures sustainable growth while maintaining high service quality and operational efficiency.

4.4.2. Profit Structure :

Monthly Profit Structure - Smart Transport Hub

Revenue Source	Revenue Generation Mechanism	Example Revenue Calculation(Monthly)
Smart Station Contracts (Universities)	Universities pay for renting or purchasing smart LED stations.	5 universities × \$3,000 = **\$15,000**
Digital Advertising on LED Screens	Businesses pay to display ads on Smart Stations.	20 ads × \$150 = **\$3,000**
Driver/Company Subscriptions	Monthly subscription to access GPS tracking system + app.	120 drivers × \$10 = **\$1,200**
Premium App Features for Students	Optional student services (alerts, seat availability, route prediction).	800 students × \$2 = **\$1,600**
Data Insights & Analytics Service	Selling transport analytics to universities/companies.	**\$2,000**
Maintenance & Service Fees	Annual/semester maintenance contracts with universities.	**\$1,500**
Total Monthly Revenue		**\$24,300**

This table simplifies the profit structure by focusing on the main revenue sources and showing the example revenue calculations for each.

- **Comparison (Revenue vs Cost)**
- **Total Fixed Monthly Cost:** ≈ **\$22,410**
- **Total Monthly Revenue: \$24,300**
- **Estimated Monthly Profit:**

\$1,890 profit / month

4.4.3. Provisional Results

This section presents the financial projection over 5 years based on the estimated costs, revenues, and growth potential of the Smart Transport Hub project.

Year	Cash Flow ($)	Discount Factor	Discounted Cash Flow	Cumulative DCF
2025	-50,000	1.00	-50,000	-50,000
2026	22,680	0.87	19,731	-30,269
2027	22,680	0.76	17,237	-13,032
2028	22,680	0.66	14,968	1,936
2029	22,680	0.57	12,927	14,863
2030	22,680	0.50	11,340	**26,203**

Net Present Value (NPV)

NPV = 26,203$ >0

Interpretation:

Since NPV > 0, the Smart Transport Hub project is **financially profitable** and **creates value**.
It generates **26,203$** in net economic benefit over 5 years.

4.4.4. Financial analysis

Net present value (NPV) over 5 years :

NPV=−I+∑(1+i)tCFt

- NVP = -I + CF1 / (1+i) + CF2 / (1+i) ^2 + CF3 / (1 +i)^3+ CF4 / (1 +i)^4+ CF5 / (1 +i)^5

  - *I* = Initial investment = 50,000
  - *CFt* = Annual cash flow = 22,680
  - *i* = Discount Rate = 15%

Net present value (NPV) over 5 years calculation

B9 =NPV(A1, A3:A7)+A2

	A	B	C
1	0.15	Discount Rate	
2	-50,000	Investment	
3	22,680	Year 1	
4	22,680	Year 2	
5	22,680	Year 3	
6	22,680	Year 4	
7	22,680	Year 5	
8			
9	NPV	######### $26,026.88	

Internal rate of return calculation

IRR is the discount rate that sets NPV = 0.

IRR ≈ 28%

Interpretation:

•IRR (28%) > Required return (15%)

•IRR is higher than bank loan interest in Lebanon

•Therefore, the project is highly attractive financially

A9 IRR

	A	B
1	0.15	Discount Rate
2	-50,000	Investment
3	22,680	Year 1
4	22,680	Year 2
5	22,680	Year 3
6	22,680	Year 4
7	22,680	Year 5
8		
9	IRR	29%

Internal rate of return calculation over the 5 years

Recovery time of the amount invested:

The recovery time is calculated by identifying the last year where the cumulative discounted cash flow is still negative and dividing the remaining unrecovered amount by the discounted cash flow of the following year. For this project, the investment is fully recovered after 3.9 years, meaning the project becomes profitable before the end of year 4.

Recovery time of the projet

	A	B	C	D
1	Year	Operating Cash Flow	Recovery Time	
2	Year 1	-50,000	-50,000	
3	Year 2	22,680	-27,320	
4	Year 3	22,680	-4,640	
5	Year 4	22,680	18,040	recovery at 3.9 year
6	Year 5	22,680	40,720	Already recovered

The project recovers the initial investment around Year 4, specifically at:

Recovery\ Time = 3 + 13,032/14,968 = 3.87 years

Profitability Index :

The profitability index is an index that attempts to identify the relationship between the costs and benefits of a proposed project

Profitability Index = (Net present value + invested amount) / invested amount

= (26,203+50,000)/50,000 = 1.52 > 1 => the project is profitable

The company can expect to cash \$ 0.52 for every dollar invested. That is, the net present value is \$ 0.52.

Bibliography

➢ **Studies, research projects, …….. :**

Lebanese Transportation Studies

5. UN-Habitat – Transport & Mobility in Lebanon
https://unhabitat.org/guide-for-mainstreaming-transport-and-mobility-in-lebanons-national-urban-policy
6. UN SDG Report – Lebanon Mobility Issues
https://ungcnlebanon.org/wp-content/uploads/2024/07/2024-SDG-11-Progress-Report.pdf
7. Al Arabiya News – Public Transport Recovery in Lebanon
https://english.alarabiya.net/News/middle-east/2025/06/03/lebanon-on-bumpy-road-to-revive-its-public-transport-

➢ **Reports :**

- Lebanese student commuting studies

  https://www.aub.edu.lb/Neighborhood/Documents/Descriptiveanalysisof2013studentstravelpatterns.pdf

- Grand View Research (2023) – Public Transportation Market Size Report

  https://www.grandviewresearch.com/horizon/outlook/public-transportation-market-size/global

- WIPO (2024) – Future of Transportation – Technology Trends

  https://www.wipo.int/edocs/pubdocs/en/wipo-pub-1055-2025-en-wipo-technology-trends-future-of-transportation.pdf

- GlobeNewsWire (2025) – Transit and Ground Passenger Transport Industry Forecast

  https://www.globenewswire.com/news-release/2025/06/16/3099997/0/en/Transit-and-Ground-Passenger-Transport-Industry-Report-2025-Market-to-Reach-1-48-Trillion-by-2034

Global Smart Mobility & ITS Market

1. Grand View Research – Smart Transportation Market
https://www.grandviewresearch.com/industry-analysis/smart-transportation-market

2. Intelligent Transportation Systems Market
   https://www.grandviewresearch.com/industry-analysis/intelligent-transportation-systems-industry
3. GlobeNewswire – Smart Transportation Market Forecast
   https://www.globenewswire.com/news-release/2024/01/23/2814001/28124/en/Global-Smart-Transportation-Market-to-2030-Adoption-of-Iot-and-Automation-Technologies-to-Enhance-Technological-Advancements.html
4. Facts & Factors — Global Smart Mobility Market size & 2030 projection (USD 243 billion by 2030) Facts and Factors

International Smart Transport Systems

8. NextBus Real-Time Transit
   https://www.nextbus.com
9. Citymapper – Mobility App
   https://citymapper.com
10. Dubai RTA Smart Bus Shelters
    https://www.rta.ae

- **Websites:**

Local Transport Apps (Competitors)

11. Bolt Lebanon
    https://bolt.eu
12. Allo Taxi
    https://allotaxi.com
13. Careem
    https://www.careem.com

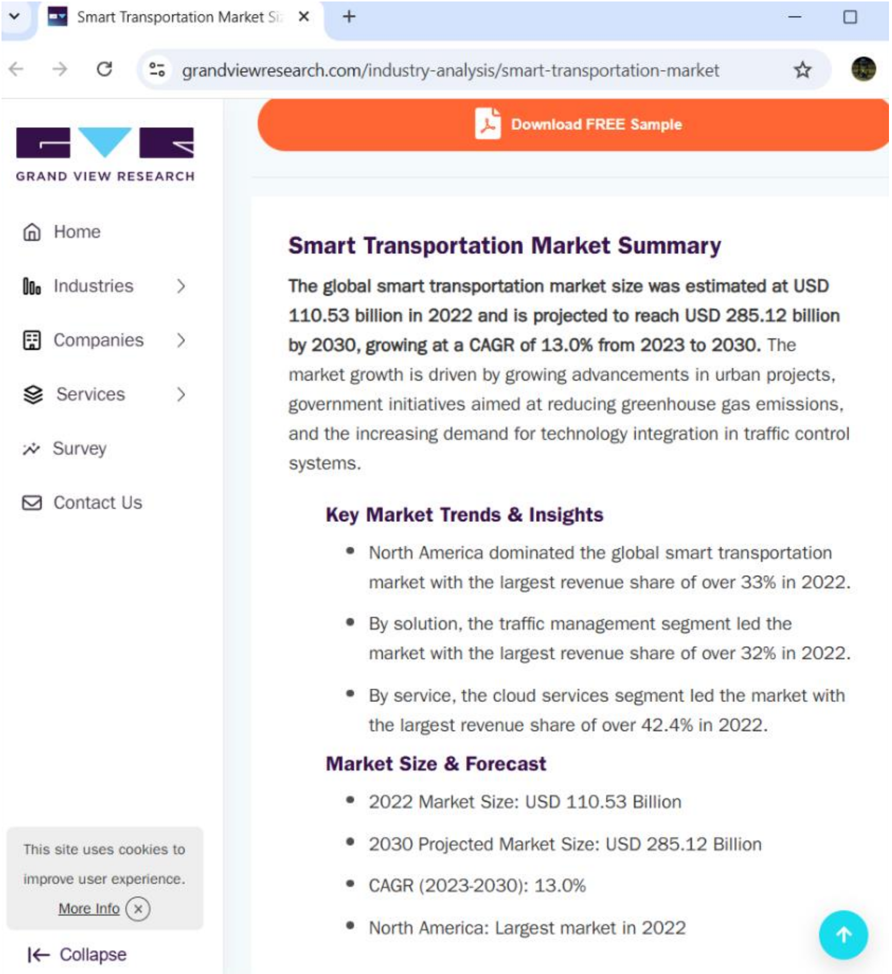
- **Our Survey :**

https://docs.google.com/forms/d/e/1FAIpQLSeMCHAuTigQh1PPZXTrebgDddzc0AUBrnTJeunxGrexPAziSA/viewform?usp=dialog

Appendices

Global Market Reports

- Grand View Research (Smart Transportation Market Report, 2023)



Smart Transportation Market Si ×

grandviewresearch.com/industry-analysis/smart-transportation-market

Download FREE Sample

GRAND VIEW RESEARCH

Home

Industries

Companies

Services

Survey

Contact Us

This site uses cookies to improve user experience. More Info

Collapse

Smart Transportation Market Summary

The global smart transportation market size was estimated at USD 110.53 billion in 2022 and is projected to reach USD 285.12 billion by 2030, growing at a CAGR of 13.0% from 2023 to 2030. The market growth is driven by growing advancements in urban projects, government initiatives aimed at reducing greenhouse gas emissions, and the increasing demand for technology integration in traffic control systems.

Key Market Trends & Insights

- North America dominated the global smart transportation market with the largest revenue share of over 33% in 2022.
- By solution, the traffic management segment led the market with the largest revenue share of over 32% in 2022.
- By service, the cloud services segment led the market with the largest revenue share of over 42.4% in 2022.

Market Size & Forecast

- 2022 Market Size: USD 110.53 Billion
- 2030 Projected Market Size: USD 285.12 Billion
- CAGR (2023-2030): 13.0%
- North America: Largest market in 2022

- GlobeNewswire (Transit & Ground Transport Market Forecast, 2025)

RESEARCH AND MARKETS
THE WORLD'S LARGEST MARKET RESEARCH STORE

Global Smart Transportation Market to 2030 - Adoption of Iot and Automation Technologies to Enhance Technological Advancements

Share

Dublin, Jan. 23, 2024 (GLOBE NEWSWIRE) -- The "Global Smart Transportation Market 2030 by Offerings and Services, Application, Transportation Mode & Region - Partner & Customer Ecosystem Competitive Index & Regional Footprints" report has been added to **ResearchAndMarkets.com's** offering.

The Smart Transportation Market size is estimated to grow from USD 120.4 Billion in 2023 to reach USD 310.65 Billion by 2030, growing at a CAGR of 14.50% during the forecast period from 2023 to 2030.

The smart transportation industry includes a wide range of technology and solutions aimed at improving transportation systems, reducing congestion, increasing safety, and increasing overall efficiency. This industry has grown in popularity as a result of rising urbanization, increased traffic congestion, environmental concerns, and the proliferation of Internet of Things (IoT) devices.

The smart transportation market strives to address the constraints and complexities of current transportation systems by leveraging cutting-edge technologies. Connected vehicles, intelligent traffic management systems, electric and autonomous vehicles, data analytics, and mobility-as-a-service (MaaS) platforms are examples of these technologies.

Company Profile

Research and Markets

Industry: Support Services

Website: https://www.researchandmarkets.com

Press Release Actions

Print
Download PDF
Subscribe via RSS
Subscribe via ATOM
Javascript

- Facts & Factors — Global Smart Mobility Market size & 2030 projection (USD 243 billion by 2030)

According to Facts & Factors, the global smart mobility market size in terms of revenue was worth around USD 48.7 billion in 2022 and is expected to reach USD 243 billion by 2030, growing at a CAGR of 19.55% from 2023 to 2030. The global Smart Mobility market is projected to grow at a significant growth rate due to a number of driving factors.

The ability to move freely and effortlessly is characterized as mobility. Mobility is essential to urban life and is the backbone of our cities. People rely on mobility to get around and do their everyday tasks. When people go from point A to point B, they desire a safe, clean, and economical mode of transportation, but they face congestion, pollution, and high traffic owing to increased urbanization. As a result, urbanization has acted as a driver for the growth of the Smart Mobility Market, which has resulted in a more accessible and multi-modal transportation system.

Lebanese Transport Studies

A2. Lebanon mobility sources

- World Bank (2023) —*Lebanon Urban Transport Report*

Implementing Agencies: Council for Development and Reconstruction, Lebanese Republic

Key Dates

Key Project Dates

Bank Approval Date: 15-Mar-2018	Effectiveness Date: 31-Jul-2019
Planned Mid Term Review Date: --	Actual Mid-Term Review Date:
Original Closing Date: 31-Dec-2023	Revised Closing Date: 31-Dec-2023

Project Development Objectives

Project Development Objective (from Project Appraisal Document)

The Project Development Objective (PDO) is to improve the speed, quality and accessibility of public transport for passengers in Greater Beirut and at the city of Beirut's northern entrance.

Has the Project Development Objective been changed since Board Approval of the Project Objective?

No

Components

Name
BRT infrastructure, fleet and systems:(Cost $230.00 M) Feeder and regular bus services and integration in urban environment:(Cost $105.00 M) Capacity building and project management:(Cost $10.00 M)

Overall Ratings

Name	Previous Rating	Current Rating
Progress towards achievement of PDO	Highly Unsatisfactory	Highly Unsatisfactory
Overall Implementation Progress (IP)	Highly Unsatisfactory	Highly Unsatisfactory
Overall Risk Rating	High	High

Implementation Status and Key Decisions

The Greater Beirut Public Transport Project was approved by the World Bank Board on March 15, 2018. The Loan Agreement was approved by the Council of Ministers on May 16, 2018 and signed on July 9, 2018. The Loan was ratified by the Parliament on July 5, 2019 and became effective on July 31, 2019.

The project faced major delays in implementation since its effectiveness. These were further exacerbated by compounded crises in Lebanon including the economic and financial crisis, the COVID-19 pandemic and lockdowns, the Port of Beirut explosion, the absence of a fully functional government for over a year, etc. The project would not have been able to meet its objectives and implement all its activities by its Closing Date on December 31, 2023, particularly considering the ongoing grave economic and financial crises and its repercussions. In this context, the World Bank processed the cancellation of the Project and notified the Borrower on July 4, 2022 of the termination of the Borrower's right to make withdrawals from the Loan Account.

Risks

Systematic Operations Risk-rating Tool

Risk Category	Rating at Approval	Previous Rating	Current Rating
Political and Governance	High	High	High
Macroeconomic	Substantial	High	High
Sector Strategies and Policies	Substantial	High	High
Technical Design of Project or Program	Substantial	High	High
Institutional Capacity for Implementation and Sustainability	Substantial	Substantial	Substantial
Fiduciary	Moderate	Moderate	Moderate
Environment and Social	Substantial	Substantial	Substantial
Stakeholders	High	High	High
Other	--	--	--
Overall	Substantial	High	High

Results

PDO Indicators by Objectives / Outcomes

Improve the speed, quality and accessibility of public transport in Greater Beirut

►Number of passengers per weekday using the formal public bus (BRT and regular buses). (Number (Thousand), Custom)

	Baseline	Actual (Previous)	Actual (Current)	End Target
Value	0.00	0.00	0.00	300.00
Date	08-Feb-2018	20-Dec-2022	01-Sep-2023	29-Dec-2023

Percentage of female ridership in the formal public bus system (BRT and regular buses) per weekday (Percentage, Custom Supplement)

	Baseline	Actual (Previous)	Actual (Current)	End Target
Value	0.00	0.00	0.00	40.00

11/1/2023 Page 2 of 6

UN-Habitat (2022) — Public mobility challenges

Guide for Mainstreaming Trans... ×

unhabitat.org/guide-for-mainstreaming-transport-and-mobility-in-lebanons-national-urban-policy?utm_source=chatgpt.com

DOWNLOAD

Share

Number of pages: 89

Guide for Mainstreaming Transport and Mobility in Lebanon's National Urban Policy

The transport sector in Lebanon is considered one of the most unsustainable in the Middle East region, due to weak governance structures and regulatory frameworks, the absence of a modern and reliable public transport system, and a car-friendly culture dominated by large old-model polluting cars. Lebanon lacks an integrated and inclusive transport system, with mostly informal public transport services and very limited infrastructure for alternative transport means.

As part of the National Urban Policy (NUP) programme in Lebanon, and following the publication of a diagnosis report, this guide proposes a set of policy orientations, recommendations and priorities to transition the transport sector in Lebanon to a sustainable future by helping to improve the state of mobility and the provision of transport services across the country. It is structured under the commonly adopted Enable-Avoid-Shift-Improve policy formulation framework for sustainable transport and mobility. The policy recommendations outline the necessary steps for designing and implementing a national transport strategy.

- Lebanese student commuting studies

https://www.aub.edu.lb/Neighborhood/Documents/Descriptiveanalysisof2013studentstravelpatterns.pdf

A travel survey was conducted in November 2013 to study the travel patterns of AUB students. The survey was carried out as part of the Neighborhood Initiative Congestion studies by faculty and students in the Civil and Environmental Engineering Department at AUB. The study investigates the commute (daily travel) to and from AUB and the walking patterns in Hamra and the neighborhood of AUB. The aim of the survey is to investigate whether students would utilize a shared-ride taxi service if it were implemented. This report presents the descriptive analyses of the sample surveyed.

Table 1 shows some descriptive statistics of the sample under study. Out of 7920 students who received an invitation to fill out the survey, 2291 have taken the survey. The below results are based upon the completed responses only.

Table 1.Sample Responses Summary

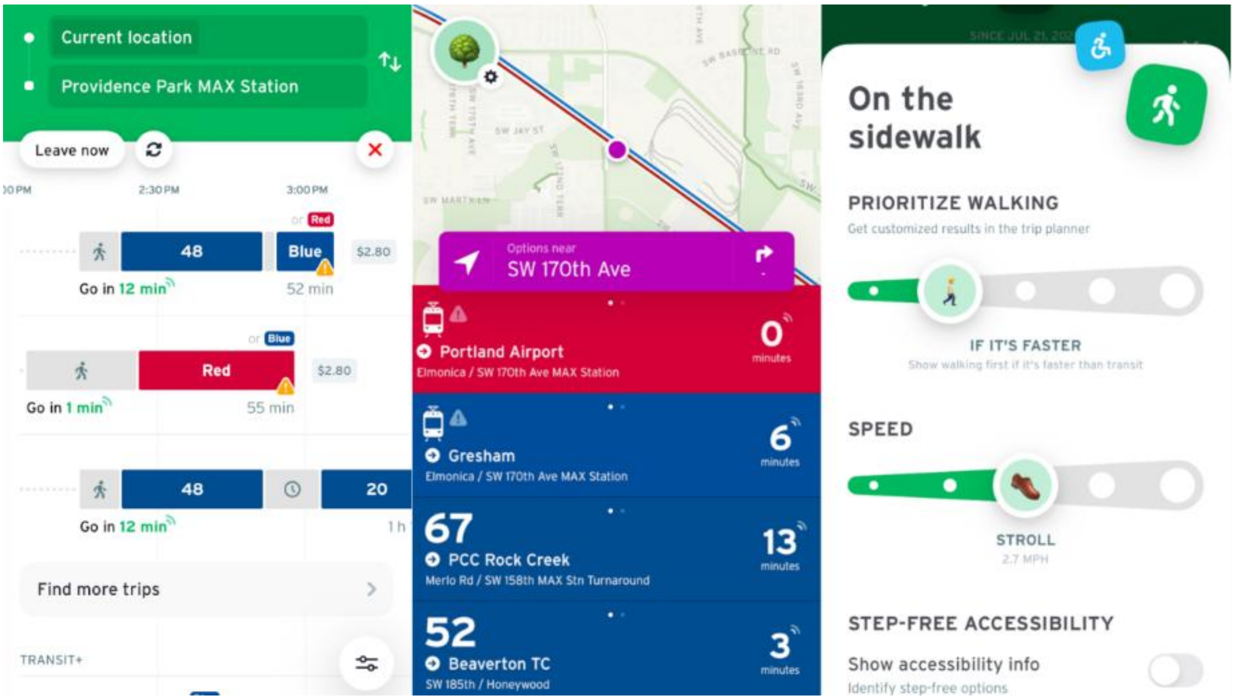
Completed Responses	Partial Responses	Total Responses	Sampling Frame Size	Response Rate
1393	898	2291	7920	17.6%

The survey was web-based (implemented via LimeSurvey) and remained active for three weeks, during which two reminder emails were sent to students.

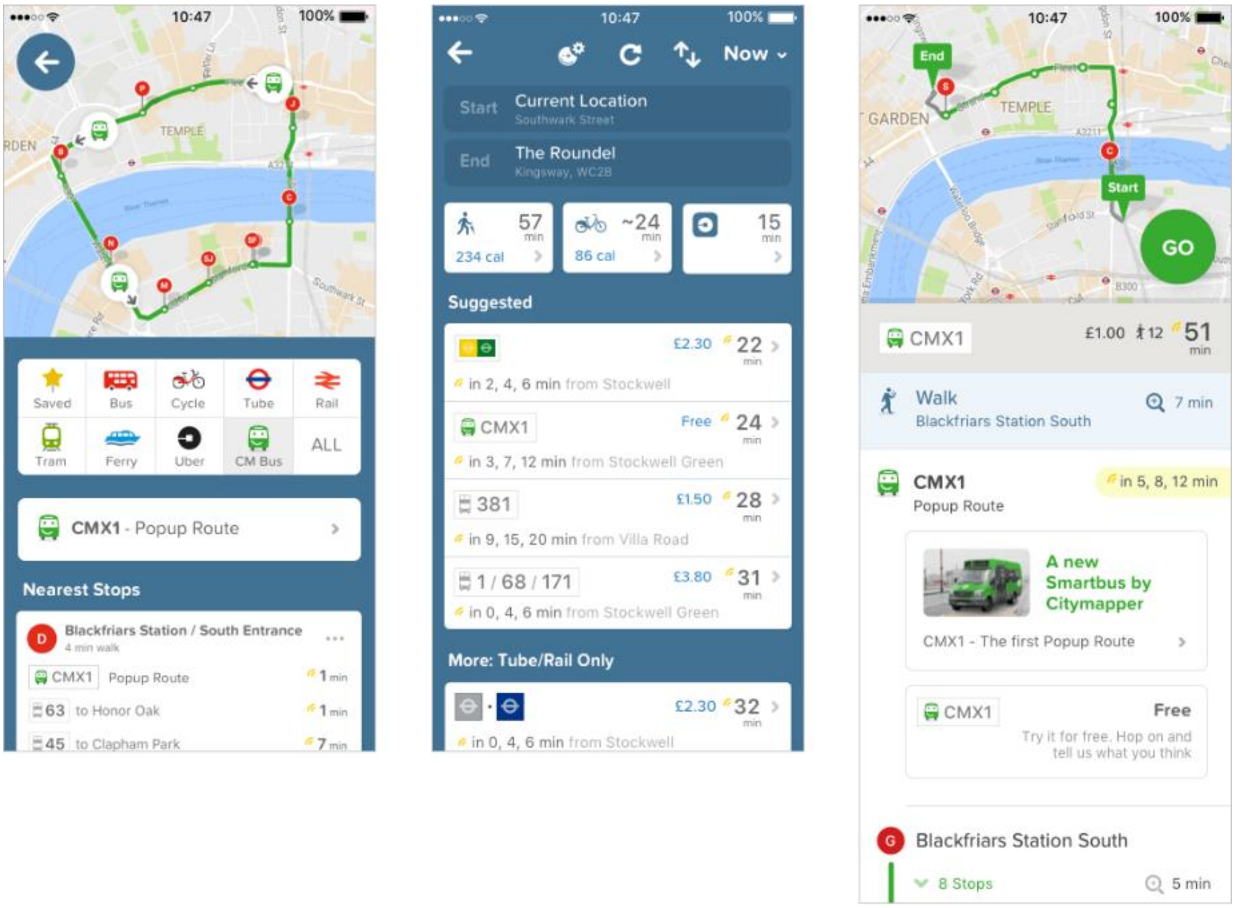
The report is divided into three main sections: socioeconomic characteristics results, commute to AUB travel patterns, and a descriptive contrast between the three travel surveys of 2007, 2010, and 2013.

Benchmark of International Systems

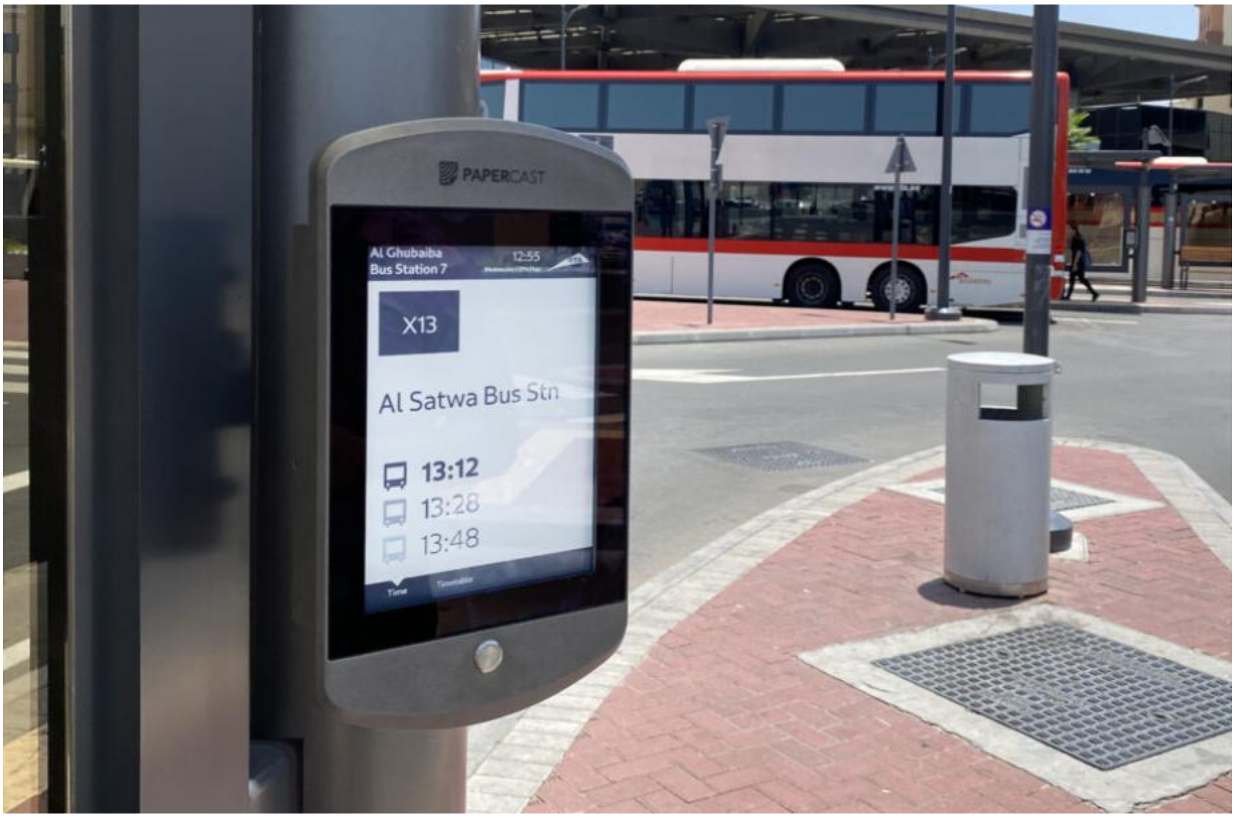
- **NextBus (USA)**



- **Citymapper (UK)**



Dubai RTA Smart Bus Stops



Images or links showing features such as:

- **NextBus** : uses GPS (satellite-based automatic vehicle location) on buses/trams to track their real-time positions. Wikipedia+2Lawrence Berkeley National Laboratory+2

  It computes and provides live estimated arrival times (ETA) for each stop — this is shown on the website, the app, or even on electronic signs at stops. sms-web.nextbus.com+2webservices.nextbus.com+2

  Users can view buses on a map, see where they are along the route, and check upcoming arrival times — which helps reduce waiting without relying on fixed schedules. nextbus-mobile.andro.io+1

- **Citymapper** : integrates multiple modes of transport (bus, subway/metro, tram, walking, cycling, etc.) to plan the best route between two points. Wikipedia+1

  For buses/trams, it now offers **real-time tracking of the vehicle on the map** — you can literally watch your bus/tram approaching your stop. Citymapper+1

  It also uses live traffic data to update travel times and ETAs dynamically, so if there is congestion or delays, your estimated arrival adjusts accordingly. www2.citymapper.com+1

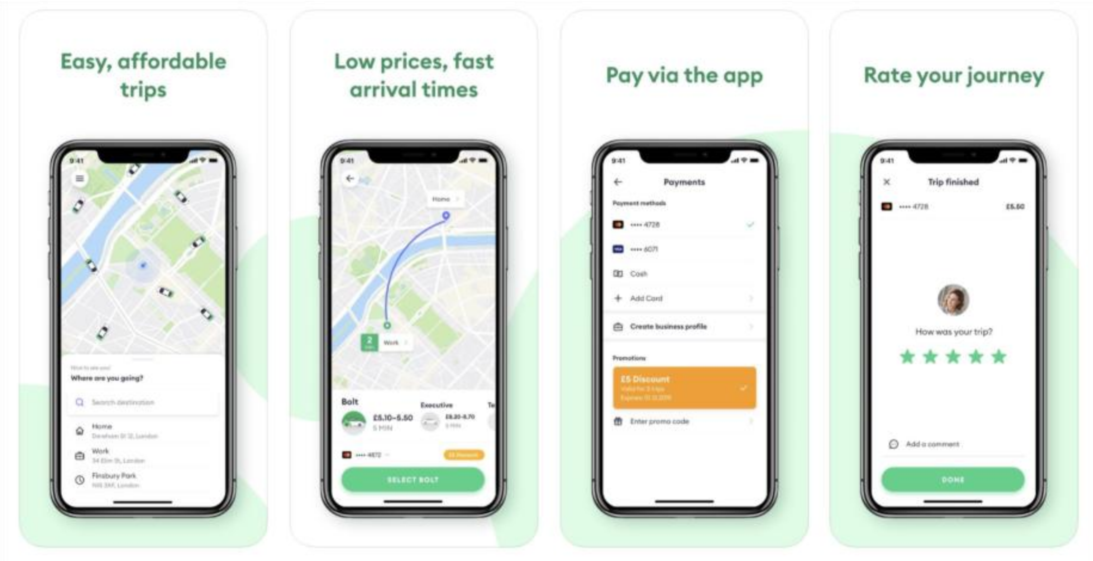
- **Dubai RTA** has installed **solar-powered real-time information screens** at many bus stops and stations — more than 150 screens planned across the city. RTA+2Zawya+2

  These screens show up-to-date bus arrival times, for the next bus and the one following it. This helps commuters know exactly when the bus is coming. Gulf News+2Gulf Today+2

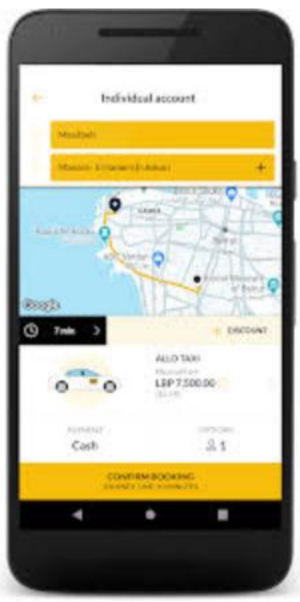
  Some of these screens are interactive/touch-screens used as journey planners — they display maps of the entire public transport network and other services. Government of Dubai Media Office+2RTA+2

Local Competitor Apps

- Bolt : Bolt works like a Uber-style taxi / ride-hailing app: you request a ride, track your driver's location, and get transported by private car. It does not function as a public transit tracker or provide anything like smart bus stops or public-transport ETA.

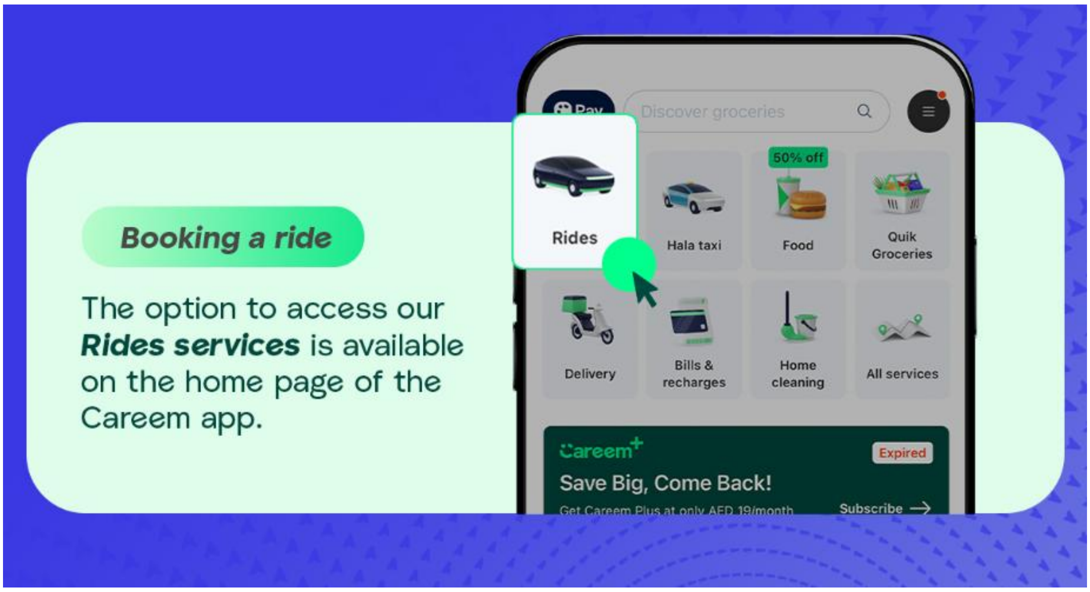


- Allo Taxi :



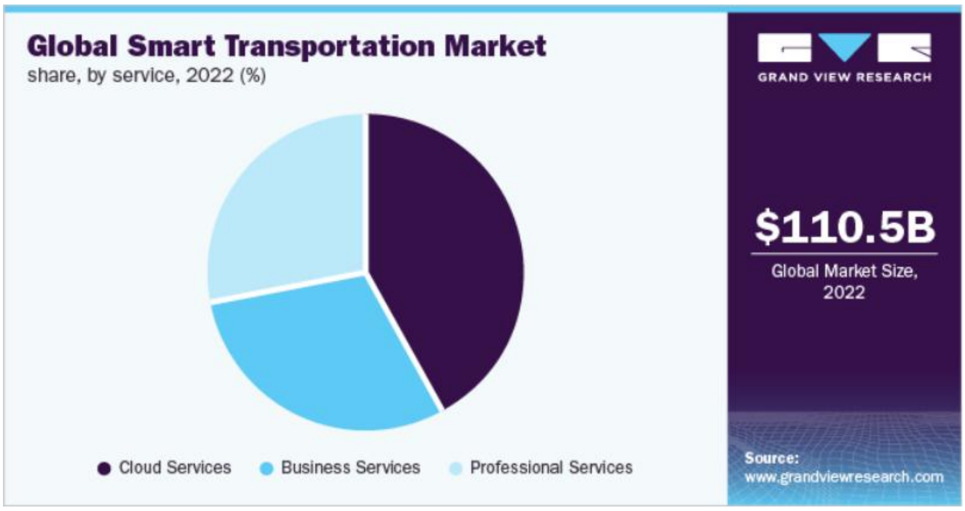
Allo Taxi works as a local taxi-booking app: good for on-demand rides when you need a car. But it does not cover public transit, buses, or bus tracking.

- Careem : Careem is a taxi / ride-hailing / mobility super-app rather than a public-transit tool. Useful when you need a car on demand, but it does *not* provide bus-tracking or smart-station services.

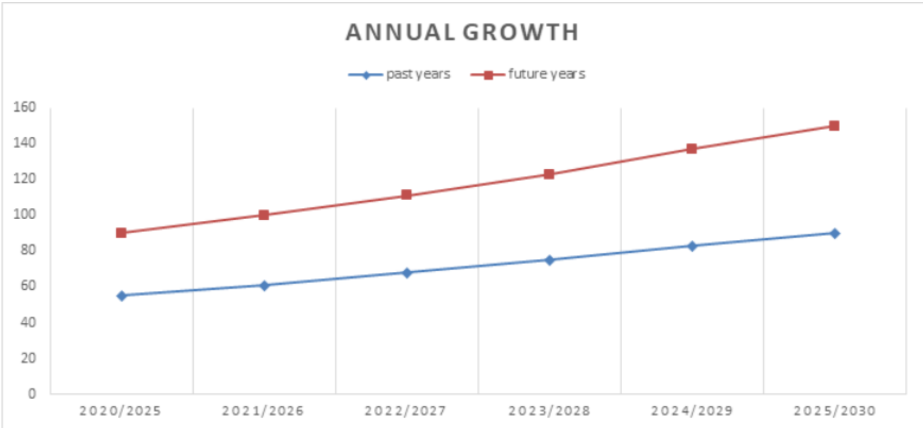


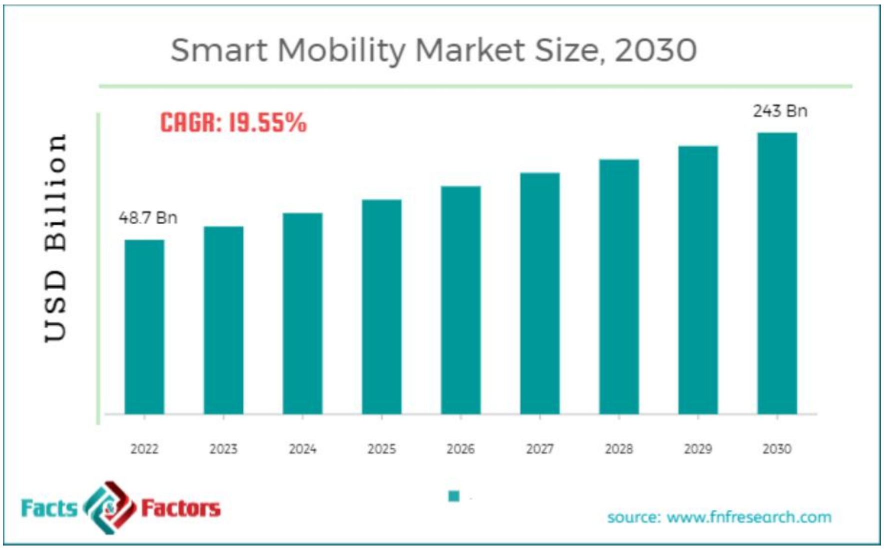
Charts & Figures Used in Research

- Global smart mobility growth graph



- Industry annual growth graph (2020–2025)





Our Survey

Student/Commuter Survey

1. Age:
    - ☐ 15-18
    - ☐ 19-22
    - ☐ 23-26
    - ☐ 27+
2. University/Institution: ____________________
3. How often do you use public transportation?
    - ☐ Daily
    - ☐ 3-4 times/week
    - ☐ Occasionally
    - ☐ Never
4. What is your usual mode of transport?
    - ☐ Bus
    - ☐ Shuttle/Van
    - ☐ Taxi
    - ☐ Other: ___________
5. How long do you usually wait for transportation?
    - ☐ Less than 10 min
    - ☐ 10-20 min
    - ☐ 20-30 min
    - ☐ More than 30 min
6. What problems do you face when commuting? (Check all that apply)
    - ☐ Long waiting time
    - ☐ Uncertain bus timing
    - ☐ Safety concerns
    - ☐ No clear updates/communication
    - ☐ Overcrowding
    - ☐ Other: ___________
7. Would you use an app that provides live tracking and accurate bus times?
    - ☐ Yes
    - ☐ Maybe
    - ☐ No
8. Which features are most important to you? (Select up to 3)
    - ☐ Live tracking
    - ☐ Notifications/alerts
    - ☐ Estimated arrival time
    - ☐ Driver rating system
    - ☐ Subscription plans for priority

- ☐ Customer support

9. How much would you pay monthly for such a service?
   - ☐ Free only
   - ☐ $1–$2
   - ☐ $3–$5
   - ☐ More than $5
10. Any suggestions or comments:

Drivers / Transport Provider Survey

1. Company/Route: ______________________
2. Years of driving experience:
   - ☐ Less than 1
   - ☐ 1–3
   - ☐ 4–7
   - ☐ 8+

3. How many trips do you make per day?
   - ☐ Less than 5
   - ☐ 5–10
   - ☐ 11–15
   - ☐ More than 15
4. Do passengers complain about waiting times?
   - ☐ Often
   - ☐ Sometimes
   - ☐ Rarely
   - ☐ Never

5. Are you currently using any tracking technology?
   - ☐ Yes
   - ☐ No
6. Would you consider using an app that helps monitor routes and schedules?
   - ☐ Yes
   - ☐ Maybe
   - ☐ No
7. What benefits would be most valuable to you? (Choose up to 3)
   - ☐ Increased passenger satisfaction
   - ☐ Better schedule management
   - ☐ Higher passenger turnout
   - ☐ Extra income opportunities
   - ☐ Reduced conflict with passengers

8. Would you be interested in partnering with a platform like rAsif?
   - ☐ Yes
   - ☐ Maybe

- ☐ No

9. If yes/maybe, how much monthly fee would be acceptable?
   - ☐ Free only
   - ☐ $2–$5
   - ☐ $6–$10
   - ☐ More than $10
10. Any comments or suggestions: